

Topic 24:

Medical Physics

24.1 Production and use of ultrasound

24.2 Production and use of X-rays

24.3 PET scanning

Remote Sensing in Medicine

Non-invasive technique

- No surgery
- No trauma
- No infection



X-ray

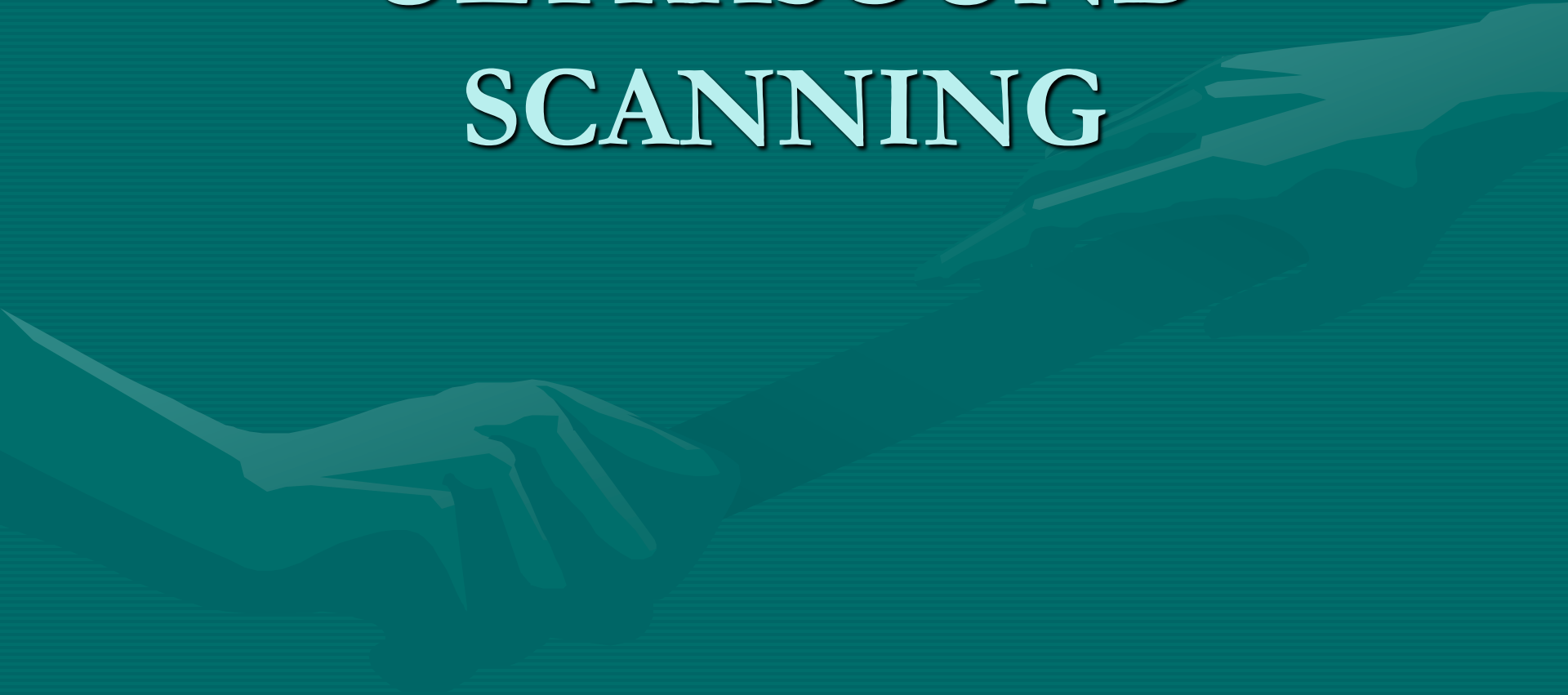
MRI

Ultra-sound

CT



ULTRASOUND SCANNING



Ultrasound Scanning

- Ultrasound scanning or **ultrasonography** is a **medical imaging technique** that **uses high frequency sound waves** and their echoes.
- The technique is **similar** to the **echolocation** used by bats, whales and dolphins, as well as **SONAR** used by submarines.



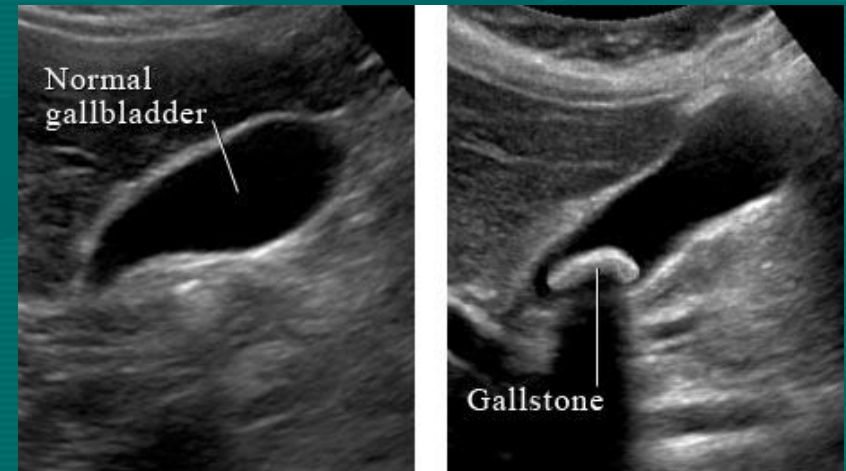
Some Applications



Monitor fetus development

Ultrasound scan is popularly used to:

- ❖ monitor **fetus development**
- ❖ analysing abnormalities of the **thyroid gland**
- ❖ **detect stones** in the gall bladder, kidneys and urinary system



Detect stones

Ultrasound Scanning Demo

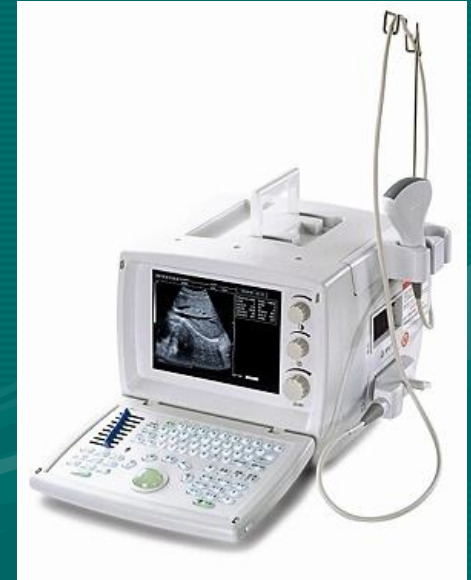


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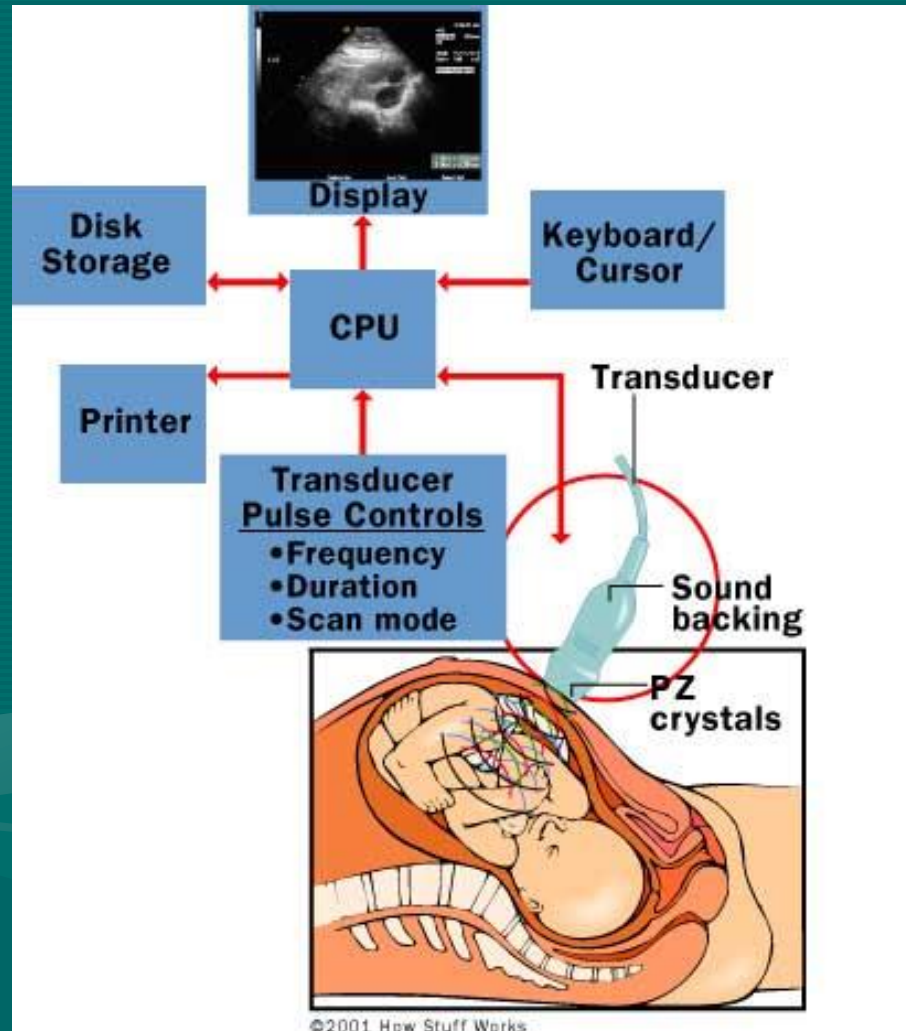
The Ultrasound Machine

A basic ultrasound machine has the following parts:

- **Transducer probe** - probe that sends and receives the sound waves
- **Transducer pulse controls** - changes the amplitude, frequency and duration of the pulses emitted from the transducer probe
- **Central processing unit (CPU)** - computer that does all of the calculations and contains the electrical power supplies for itself and the transducer probe
- **Display** - displays the image from the ultrasound data processed by the CPU
- **Other peripherals:**
 - **Keyboard/cursor** - inputs data and takes measurements from the display
 - **Disk storage device** (hard, floppy, CD) - stores the acquired images
 - **Printer** - prints the image from the displayed data

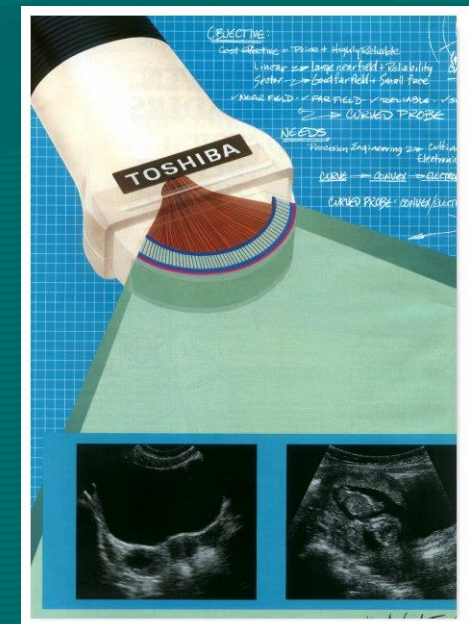
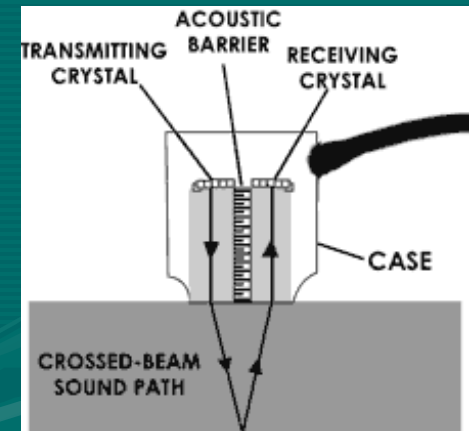


Block Diagram

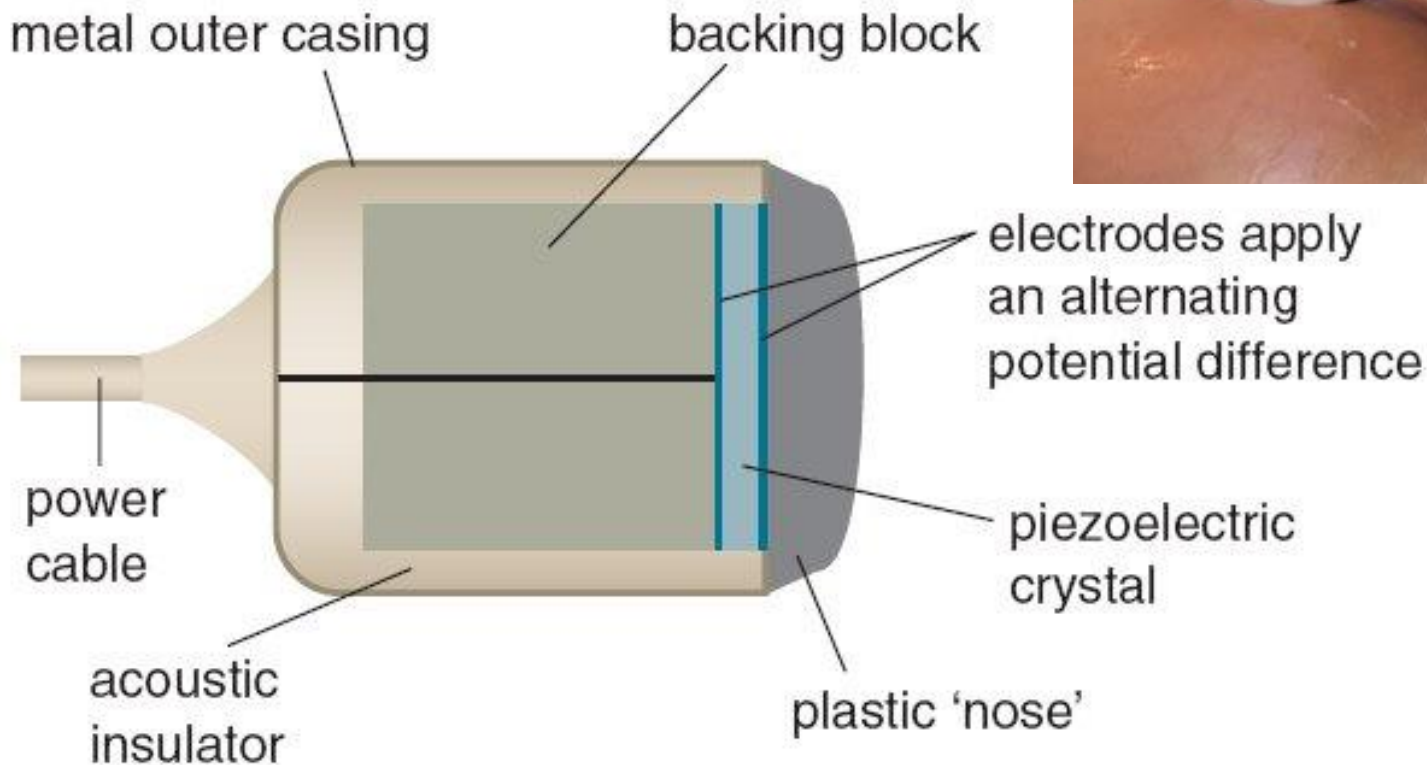


The Process

- The **transducer generates sound waves** by using electrical potential to vibrate crystals within the transducer.
- The transducer **sends out sound waves** in pulses.
- Between pulses, it **listens for echoes** that are **returning from structures within the body**.
- The ultrasound system **produces an image based on** the **time** it takes for sound waves to return to the transducer and the **strength** of those sound waves.
 - Sound waves that take **longer to return** will show up as **farther away** in the image.
 - **Stronger** sound waves will show up as **brighter** objects.



Structure of the Transducer Probe

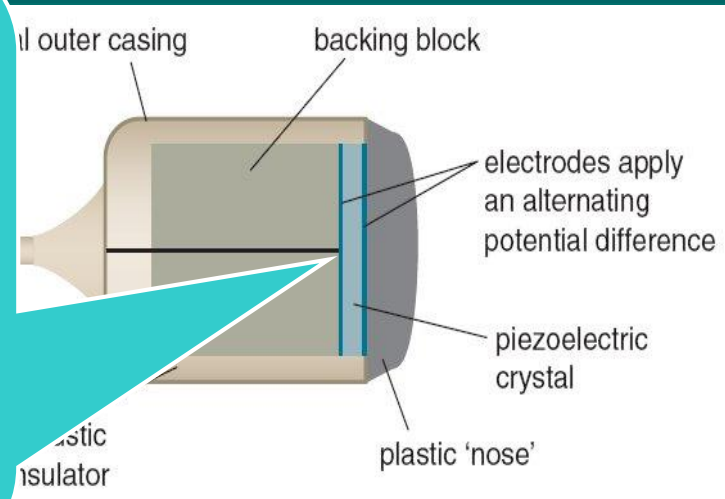
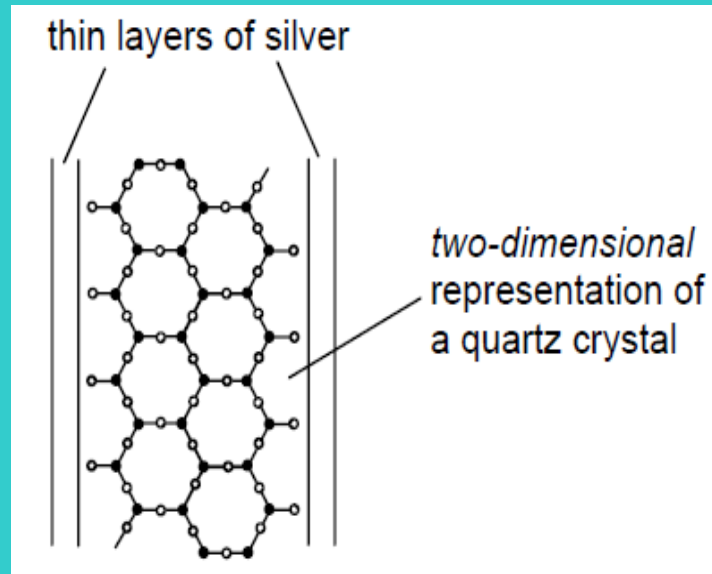


The Transducer Probe



- The **transducer probe** is the **main part of the ultrasound machine**.
- The transducer probe **makes the sound waves and receives the echoes**. It is, so to speak, the mouth and ears of the ultrasound machine.
- The transducer probe generates and receives sound waves using a principle called the **piezoelectric** (pressure electricity) **effect**, which was discovered by Pierre and Jacques Curie in 1880.
- In the probe, there are one or more **quartz crystals** called piezoelectric crystals.
- **When a potential difference is applied** between these crystals, they **change shape rapidly**.
- The rapid shape changes, or vibrations, of the crystals **produce sound waves** that travel outward.
- **Conversely, when sound or pressure waves hit the crystals, they emit electrical pulses**.
- Therefore, the **same crystals** can be used to **send and receive sound waves**.
- The probe also has a **sound absorbing substance** to eliminate back reflections from the probe itself, and an acoustic lens to help focus the emitted sound waves.

The Piezoelectric Transducer

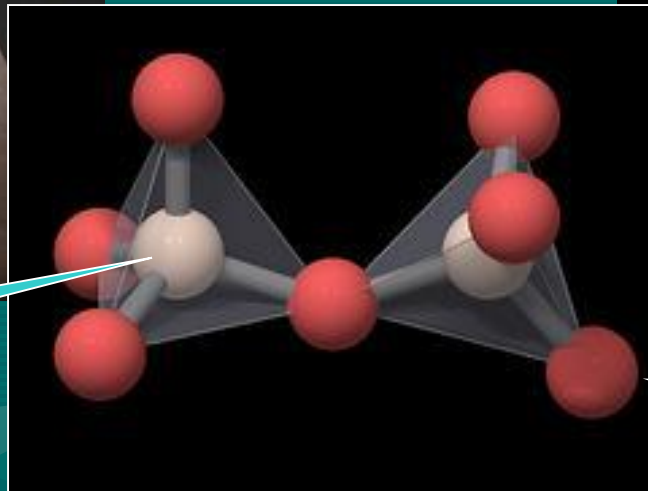


The **piezoelectric transducer** is made up of a piece of **quartz crystal** with its **two opposite sides coated with thin layers of silver** to act as electrical contacts.

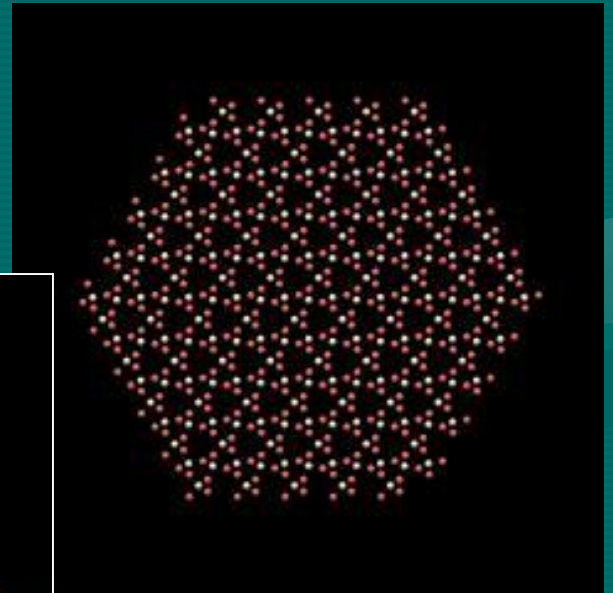
The Quartz Crystal



positively-charged
silicon ion

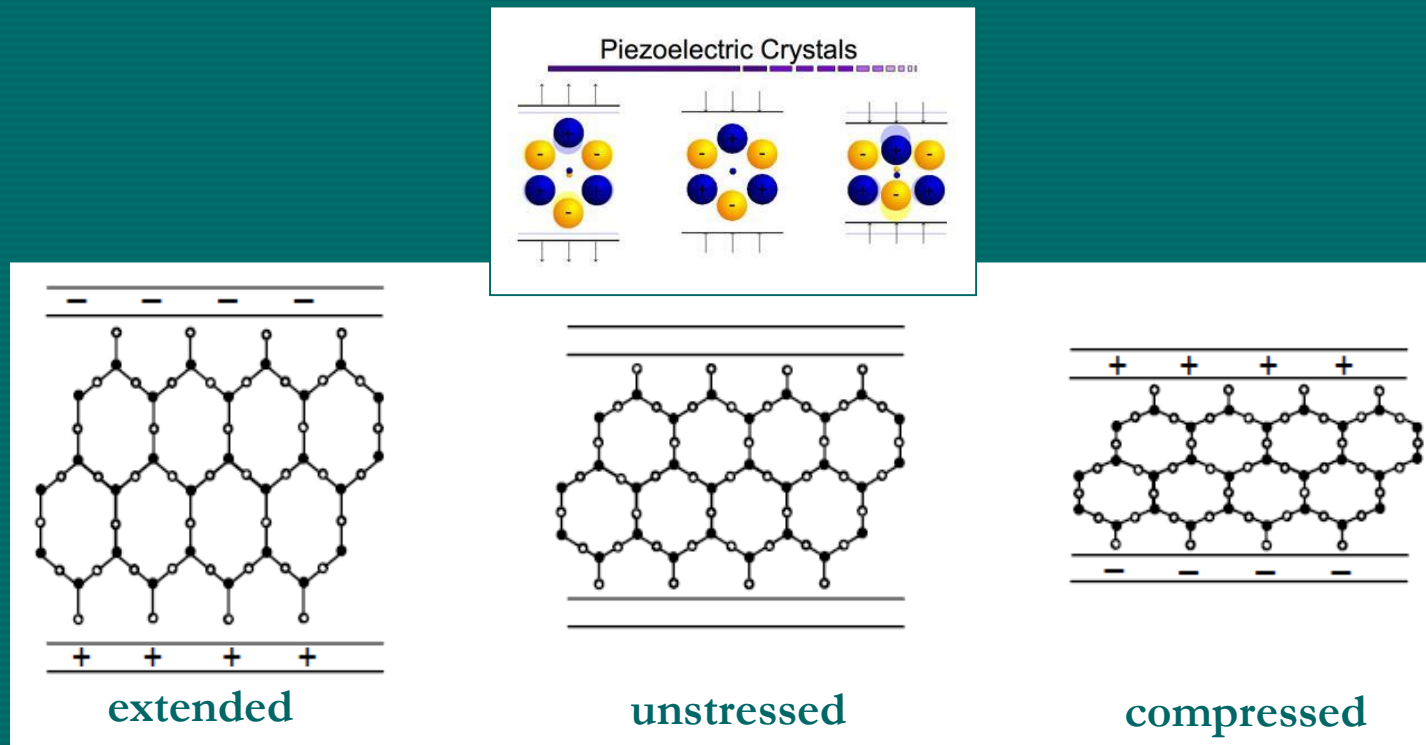


negatively-charged
oxygen ion



Quartz is made up of a large number of repeating **tetrahedral silicate units**. The positions of the oxygen links are **not rigidly fixed** in these units, or lattices, and since the oxygen ions are negatively charged, **movement can be encouraged by applying an electric field**.

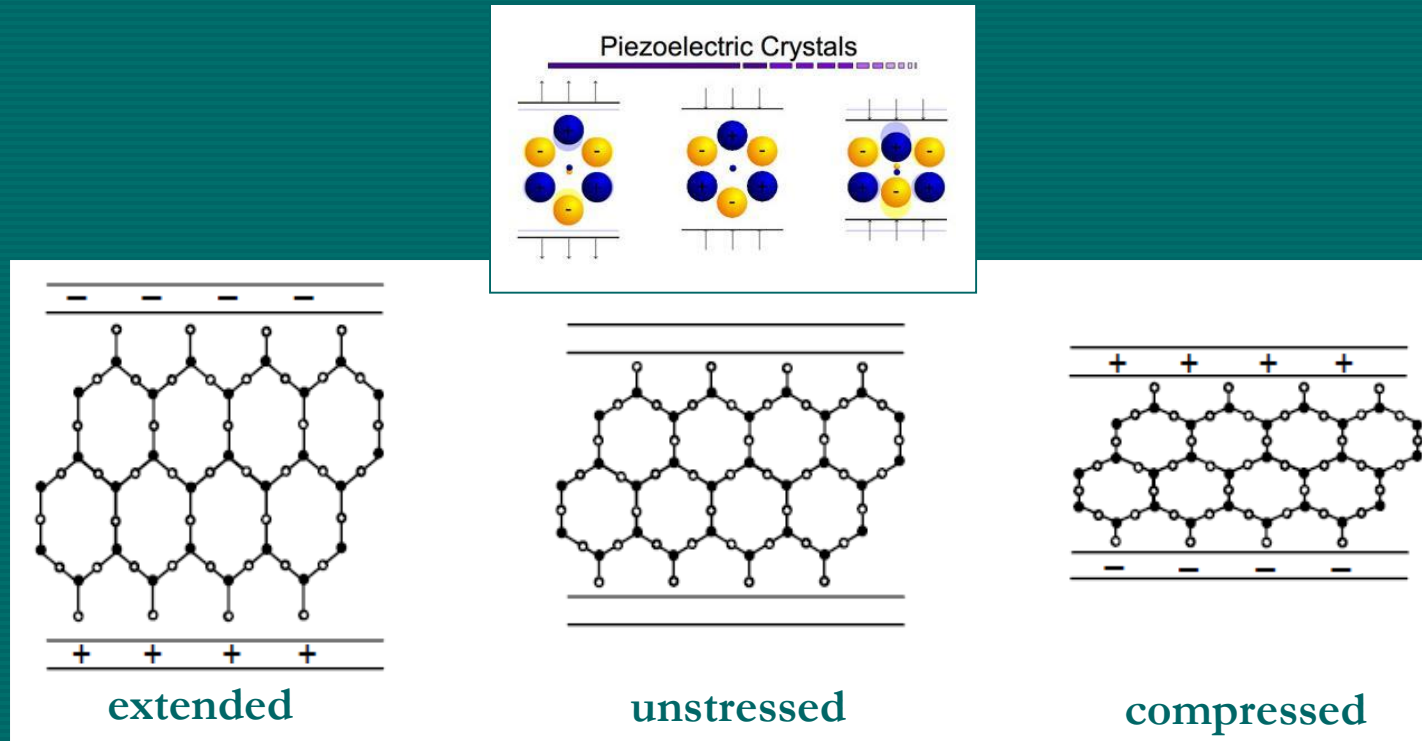
The Production of Sound Waves (1)



When the crystal is unstressed, the centres of charge of the positive and the negative ions bound in the lattice of the piezo-electric crystal coincide, so their effects are neutralised.

If a constant voltage is applied across the electrodes, the positive silicon ions are attracted towards the cathode and the negative oxygen ions towards the anode. This causes distortion of the silicate units. Depending on the polarity of the applied voltage, the crystal becomes either thinner or thicker as a result of the altered charge distribution.

The Production of Sound Waves (2)

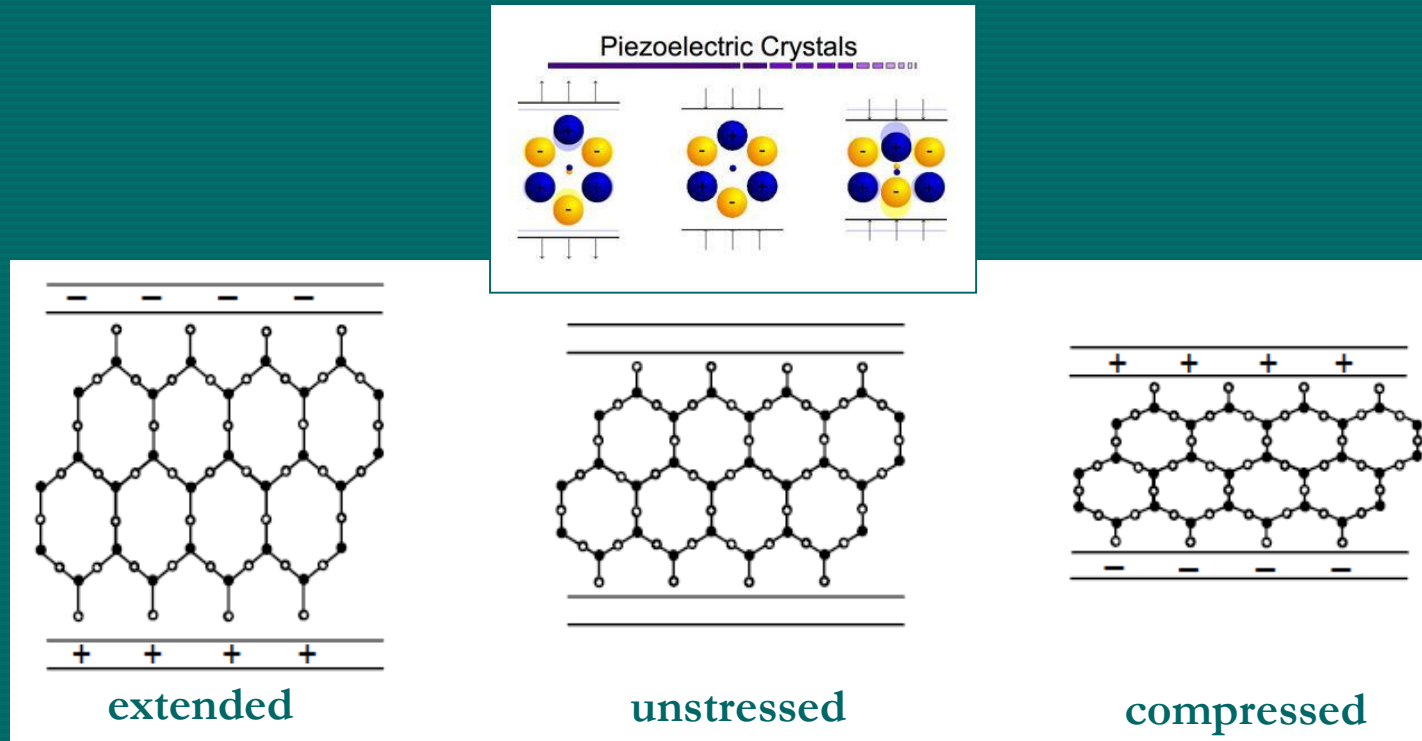


An alternating voltage applied across the silver electrodes will set up mechanical vibrations in the crystal.

If the frequency of the applied voltage is the same as the natural frequency of vibration of the crystal, resonance occurs and the oscillations have maximum amplitude.

The dimensions of the crystal can be such that the oscillations are in the ultrasonic range (i.e. greater than 20 kHz), thus producing ultrasonic waves in the surrounding

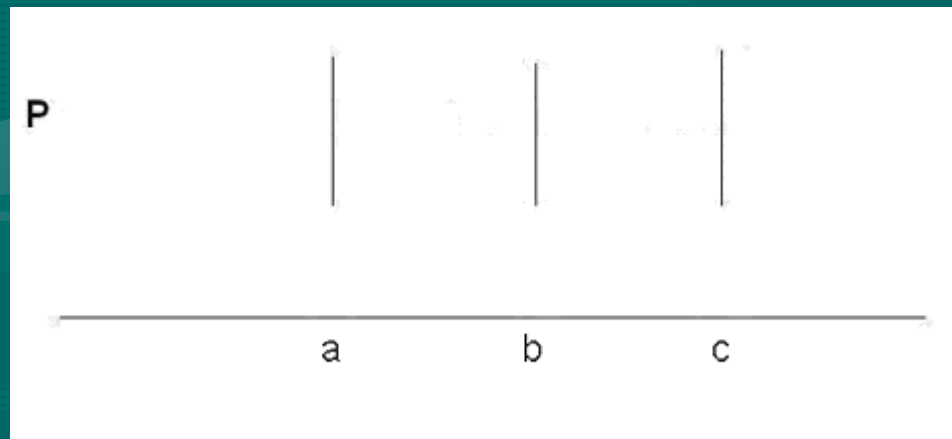
The Production of Electrical Pulses



- ❖ Ultrasonic transducers can also be used as receivers.
- ❖ When an ultrasonic wave is incident on an unstressed piezo-electric crystal, the pressure variations alter the positions of positive and negative ions within the crystal.
- ❖ This induces opposite charges on the silver electrodes, producing a potential difference between them.
- ❖ This varying potential difference can then be amplified and processed.

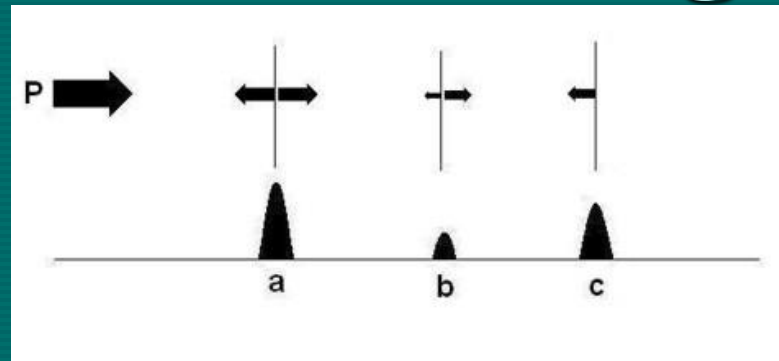
Ultrasound Imaging

Ultrasound imaging is performed by emitting a pulse, which is partly reflected from a boundary between two tissue structures, and partially transmitted. The reflection depends on the difference in impedance of the two tissues.



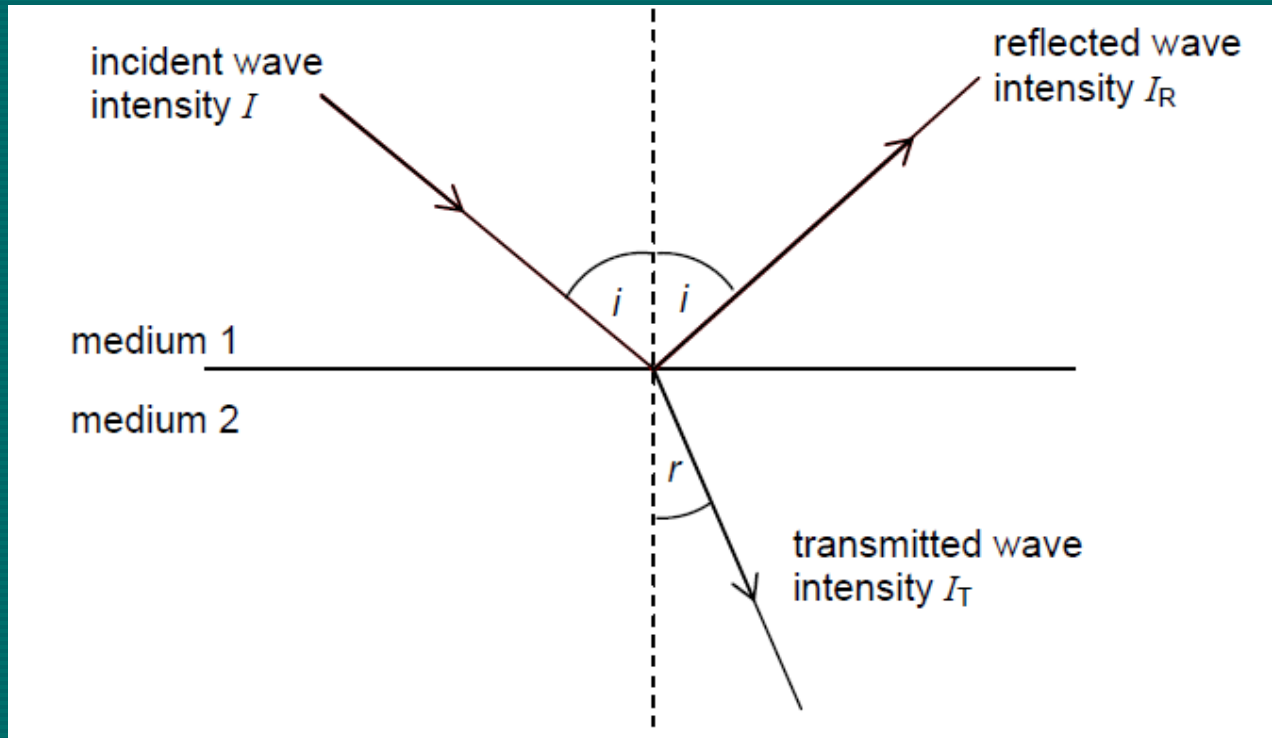
Reflection at tissue boundaries a, b and c.

Ultrasound Imaging



- An ultrasound pulse is emitted from the probe P.
- Part of the pulse energy is reflected from the scatterer a, the rest is transmitted.
- Part of the energy transmitted at a is reflected from b and the rest transmitted.
- The energy transmitted at b will be reflected when it hits c.
- When the pulse returns to P, the reflected pulse gives information of two measurements:
 - The amplitude of the reflected signal,
 - and the time it takes returning.
- The time taken is dependent on the distance from the probe (twice the time the sound uses to travel the distance between the transmitter and the reflector, as the sound travels back and forth).
- The amplitude indicate the amount of energy being reflected from each point.
 - Thus, the incoming (incident) pulse at a has the full amplitude of P.
 - At b, the incoming pulse is the pulse transmitted through a.
 - At c, the incident pulse is the transmitted pulse from b.(In both cases minus further attenuation in the interval.)

Laws of Reflection & Refraction



Ultrasound obeys the same laws of reflection and refraction at boundaries as audible sound and light.

For an incident intensity I , reflected intensity I_R and transmitted intensity I_T , then from energy considerations,

$$I = I_R + I_T.$$

Acoustic Impedance

- The relative magnitudes of the reflected and transmitted intensities depend not only on the angle of incidence but also on the **acoustic impedance** of the two media.
- The specific acoustic impedance Z of a medium is the **speed of sound** in the material **multiplies** the **density**:
$$Z = \rho \times c$$
- The ratio I_R / I is known as the **intensity reflection coefficient** for the boundary and is usually given the symbol α

$$\alpha = \frac{I_R}{I} = \frac{(Z_2 - Z_1)^2}{(Z_2 + Z_1)^2}$$

Velocity, Impedance & Absorption Coefficient

The sound velocity in a given material is constant (at a given temperature), but varies in different materials:

Material	Velocity (m/s)
Air	330
Water	1497
Metal	3000 - 6000
Fat	1440
Blood	1570
Soft tissue	1540

medium	$Z = \rho c / \text{kg m}^{-2} \text{ s}^{-1}$
air	430
quartz	1.52×10^7
water	1.50×10^6
blood	1.59×10^6
fat	1.38×10^6
muscle	1.70×10^6
soft tissue	1.63×10^6
bone	$(5.6 - 7.8) \times 10^6$

medium	absorption coefficient / m^{-1}
air	120
water	0.02
muscle	23
bone	130

Example 1

Medium	Speed/ ms ⁻¹	Z / kg m ⁻² s ⁻¹
Air	330	430
Blood	1600	1.6×10^6
Bone	4100	$5.6 \times 10^6 - 7.8 \times 10^6$
Fat	1500	1.4×10^6
Muscle	1600	1.7×10^6
Soft tissue	1600	1.6×10^6
Water	1500	1.5×10^6

Using the data in the table, calculate the intensity reflection coefficient for a parallel beam of ultrasound incident normally on the boundary between:

- (a) Air and soft tissue
- (b) Muscle and bone that has a specific acoustic impedance of $6.5 \times 10^6 \text{ kg m}^{-2} \text{ s}^{-1}$

Solution:

(a)

$$\alpha = \frac{(Z_2 - Z_1)^2}{(Z_2 + Z_1)^2} = \frac{(1.6 \times 10^6 - 430)^2}{(1.6 \times 10^6 + 430)^2} = 0.999$$

(b)

$$\alpha = \frac{(Z_2 - Z_1)^2}{(Z_2 + Z_1)^2} = \frac{(6.5 \times 10^6 - 1.7 \times 10^6)^2}{(6.5 \times 10^6 + 1.7 \times 10^6)^2} = 0.343$$

Example 2

Medium	Speed/ ms ⁻¹	Z / kg m ⁻² s ⁻¹
Air	330	430
Blood	1600	1.6×10^6
Bone	4100	$5.6 \times 10^6 - 7.8 \times 10^6$
Fat	1500	1.4×10^6
Muscle	1600	1.7×10^6
Soft tissue	1600	1.6×10^6
Water	1500	1.5×10^6

Using the data in the table:

- (a) Suggest why, although the speed of ultrasound in blood and muscle is approximately the same, the specific acoustic impedance is different.
- (b) Calculate the intensity reflection coefficient for a parallel beam of ultrasound incident normally on the boundary between fat and muscle.

Solution:

- (a) Their density is different, muscle has a higher density and hence a higher specific acoustic impedance.

(b)

$$\alpha = \frac{(Z_2 - Z_1)^2}{(Z_2 + Z_1)^2} = \frac{(1.7 \times 10^6 - 1.4 \times 10^6)^2}{(1.7 \times 10^6 + 1.4 \times 10^6)^2} = 9.4 \times 10^{-3}$$

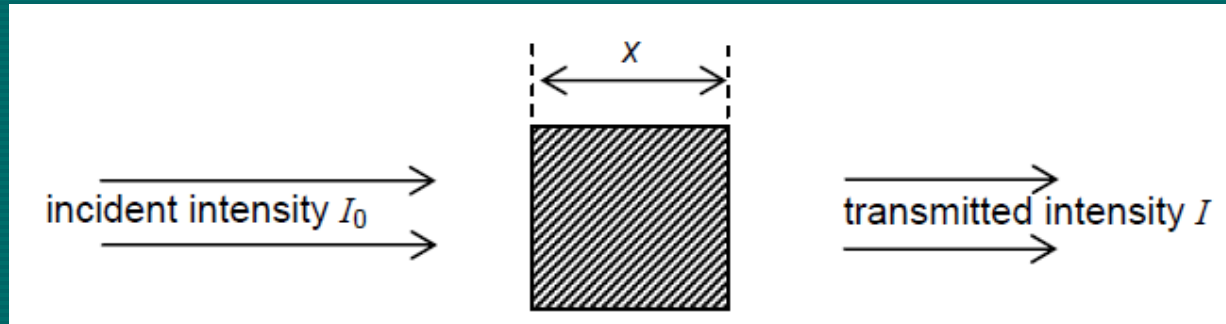
Use of Gel



When in use, the transducer is placed in contact with the skin, with a **gel** acting as a coupling medium.

The gel **reduces the size of the impedance change between boundaries** at the skin and thus **reduces reflection at the skin**, enabling more waves to enter the body.

Absorption of Energy



- A second factor that affects the intensity of ultrasonic waves passing through a medium is **absorption**.
- As a wave travels through a medium, energy is absorbed by the medium and the **intensity of a parallel beam decreases exponentially**. The **temperature of the medium rises**.
- The **intensity I** of the beam after passing through the medium is related to the incident intensity by the expression

$$I = I_0 e^{-kx}$$

where k is a constant for the medium referred to as the **absorption coefficient**. This coefficient is **dependent on the frequency** of the ultrasound.

Example 3

medium	absorption coefficient / m ⁻¹
air	120
water	0.02
muscle	23
bone	130

A parallel beam of ultrasound is incident on the surface of a muscle and passes through a thickness of 3.5 cm of the muscle. It is then reflected at the surface of a bone and returns through the muscle to its surface. Using data from the tables, calculate the fraction of the incident intensity that arrives at the surface of the muscle.

Solution:

The beam passes through a total thickness of 7.0 cm of muscle.

Therefore, $I = I_0 e^{-kx} = I_0 e^{-0.23 \times 7.0} = 0.20 I_0$

It is obtained from the previous example,

Fraction of sound reflected at the muscle-bone interface = 0.34

Therefore fraction received back at surface = $0.34 \times 0.20 = 0.068 = 1/15$

Practice 4

medium	absorption coefficient / m^{-1}
air	120
water	0.02
muscle	23
bone	130

A parallel beam of ultrasound passes through a thickness of 4.0 cm of muscle. It is then incident normally on a bone having a specific acoustic impedance of $6.4 \times 10^6 \text{ kg m}^{-2} \text{ s}^{-1}$. The bone is 1.5 cm thick. Using data from the table, calculate the fraction of the incident intensity that is transmitted through the muscle and bone.

Answer:

Significance of Absorption

Absorption has raised **two important considerations**:

- The heating of the tissue is the safety limitation on ultrasound equipment. **The absorbed energy has to remain within limits that does not heat the tissue to dangerous temperatures.** The absorption can be calculated, and in commercial equipment today, the limitation is imposed in the limitation of the total energy that can be transmitted.
- The absorption is **the limiting factor for the depth penetration of the beam**, i.e. the depth to which the beam can be transmitted.

A side benefit:

- The heating effect caused by ultrasound of suitable frequencies is, in fact, **used in physiotherapy** to assist recovery from sprained joints.

Factors Affecting Absorption

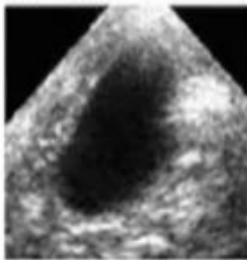
Absorption is dependent on:

- The **density of the tissue**. **The higher the density, the more absorption.**
 - Thus the attenuation is:
 - fluid < fat < muscle < fibrous tissue < calcifications and bone.
- The **frequency of the ultrasound beam**. **The higher the frequency, the more absorption.**
 - Thus, the desired depth to be imaged, sets the limit for how high frequency that can be used.
 - As can be seen, penetration might be increased by increasing the transmitted energy, but this would increase the total absorbed energy as well, which has to stay below the safety limits.
 - In practice, a lower frequency (e.g, 3.5 MHz) is used for deeper penetration and a higher frequency (e.g 7.5 MHz) for superficial areas. (Higher frequency used will give image of better resolution)

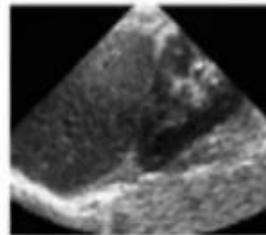
Images

signostics

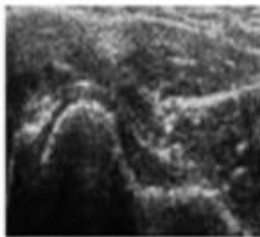
The Basic Principles of Ultrasound Image Interpretation



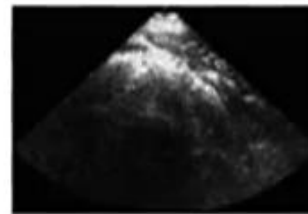
Fluids appears Black (anechoic) due to the lack of interfaces.



Tissues and Organs have their own Characteristic Shape & Echopattern
e.g. Liver homogeneous
Kidney with Cortex



Bones and Calculi cast dark acoustic shadows. Sound does not penetrate deep to the visible surface of the bone or calculus



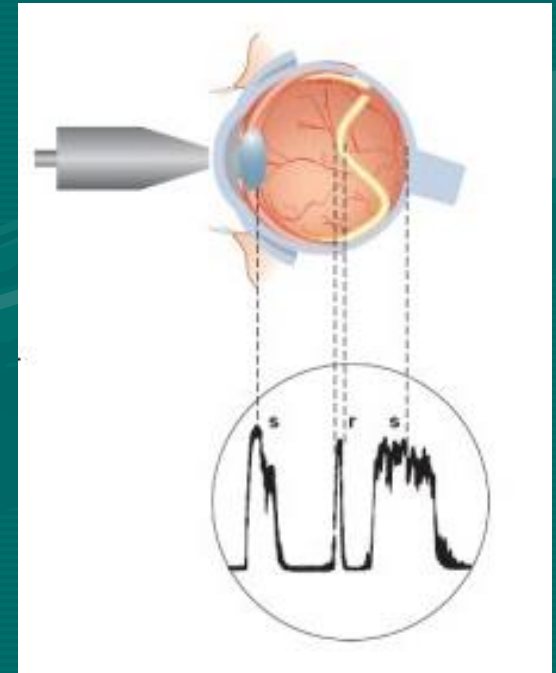
Gas casts bright acoustic shadows due to resonance of the gas bubbles
e.g. Lungs, Bowel.



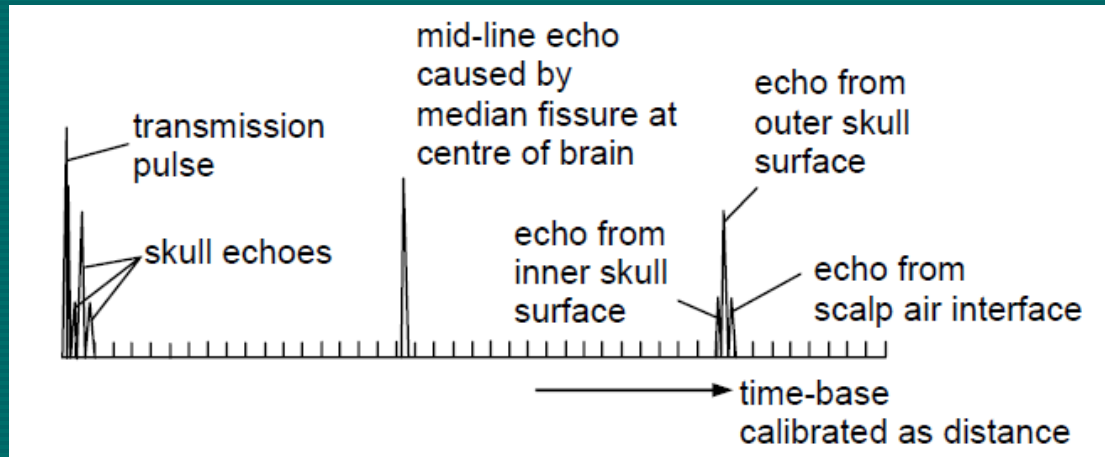
Personal Ultrasound

A-scans

- A-scans are used to measure distances. A transducer emits an ultrasonic pulse and the time taken for the pulse to bounce off an object and come back is graphed in order to determine how far away the object is. A-scans only give one-dimensional information and therefore are not useful for imaging.

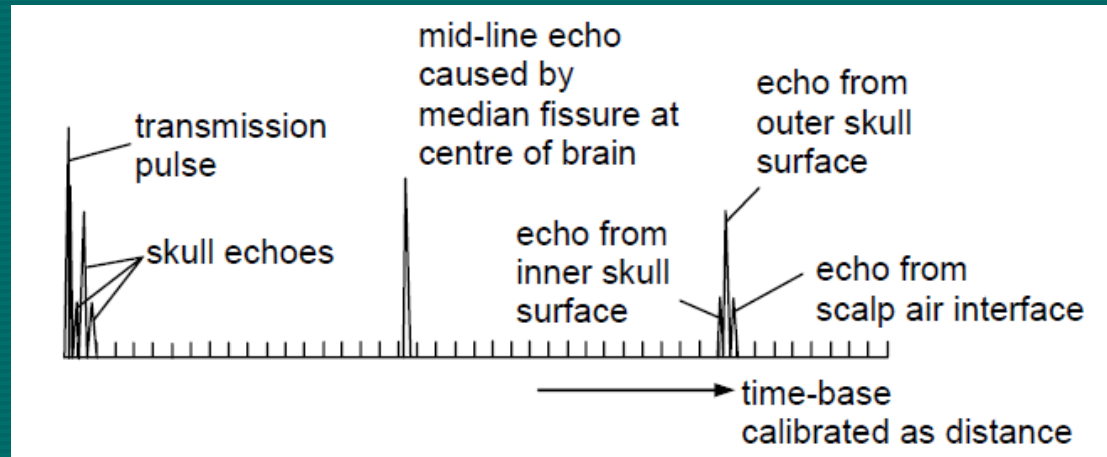


The A-scan System (1)



- The **A-scan system** basically **measures the distance of different boundaries from the transducer**, with the **transducer held in one position**.
- A **short burst of ultrasound is transmitted to the body** through the coupling medium.
- **At each boundary** between different media in the body, **some ultrasound is reflected** and **some is transmitted**.
- The **reflected pulse is picked up** by the transducer which now acts as a receiver. The signal is **amplified** and **displayed** on a cathode-ray oscilloscope (c.r.o.).
- The reflected pulse also meets boundaries as it returns to the transducer. This causes **some of the energy of the reflected pulse to be lost** and **energy is also lost due to absorption** in the media.

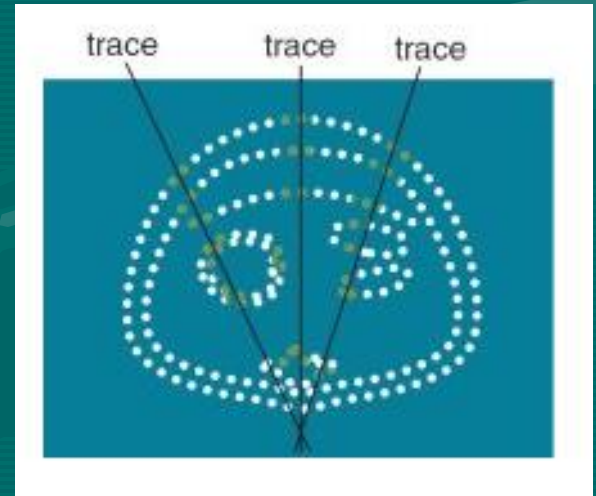
The A-scan System (2)



- Consequently, **echoes from deeper in the body tend to be of lower intensity**. To compensate for this, the later an echo is received at the transducer, **the more it is amplified** before display on the c.r.o.
- **A vertical line appears on the screen each time an echo is received**.
- The **time-base** on the X-plates **is adjusted** so that all of the reflections are seen on the screen for one scan (pulse).
- The **distance between boundaries can be calculated** if the speed of ultrasound in the various media is known. An example of an A-scan for the brain is shown in diagram above.

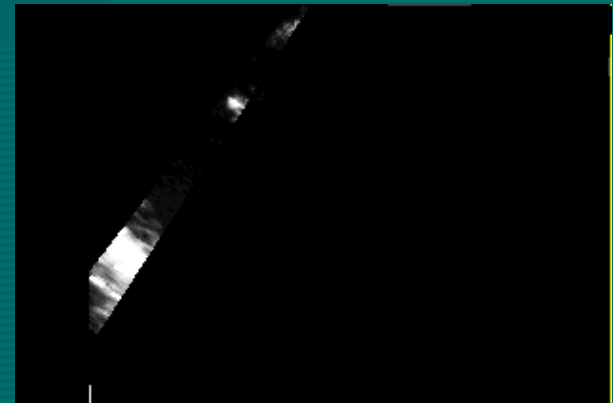
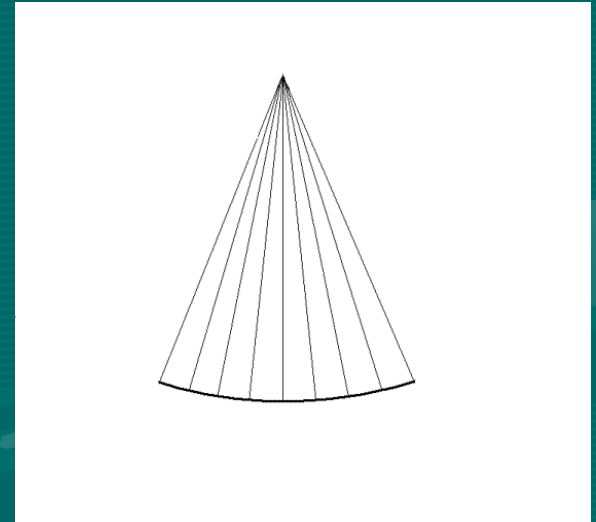
B-scans

- B-scans can be used to take an image of a cross-section through the body. The transducer is swept across the area and the time taken for pulses to return is used to determine distances, which are plotted as a series of dots on the image. B-Scans will give two-dimensional information about the cross-section.

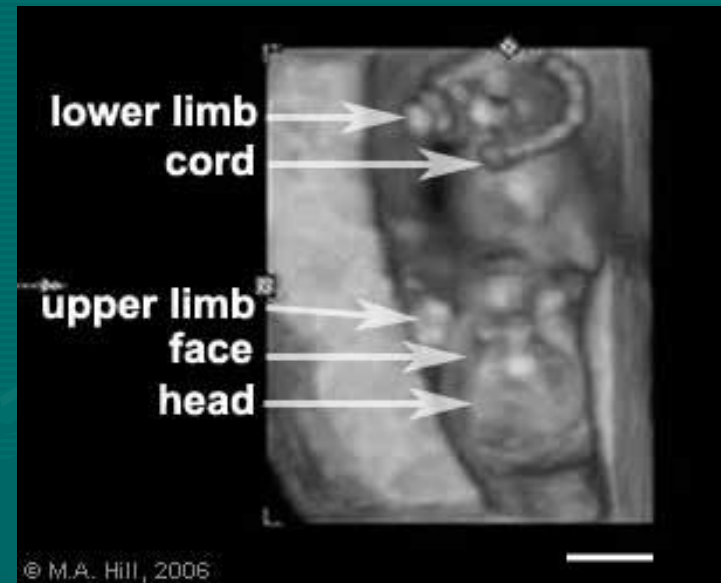
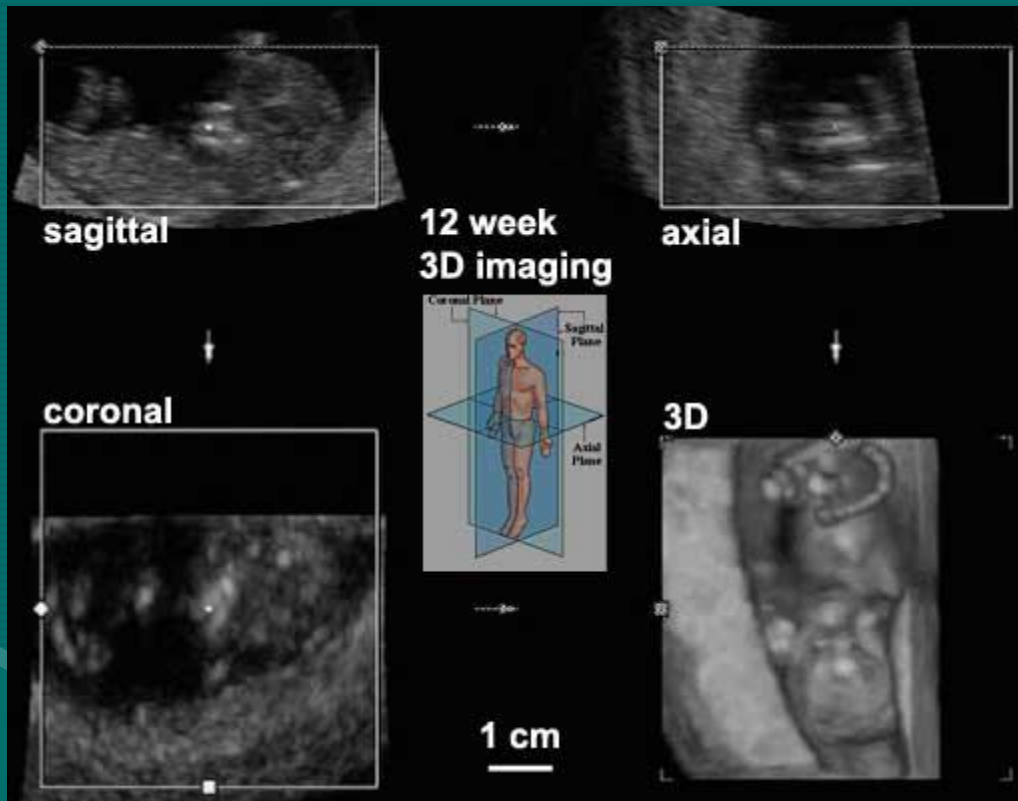


The B-scan System

- The B-scan technique basically combines a series of A-scans, taken from a range of different angles, to form a two-dimensional picture. As before, each A-scan corresponds to a single ultrasound pulse being emitted by the transducer and producing a series of reflected pulses from boundaries within the body.
- The ultrasound probe for a B-scan consists of a series of small crystals, each having a slightly different orientation. The signals received from the crystals in the probe are processed by a computer. Each reflected pulse is shown as a bright spot in the correct orientation of the crystal on the screen of a c.r.o.
- Consequently, the completed pattern of spots from all the crystals in the probe builds up into a two-dimensional representation of the boundary positions in the body being scanned. This image may be photographed or stored in the computer memory.



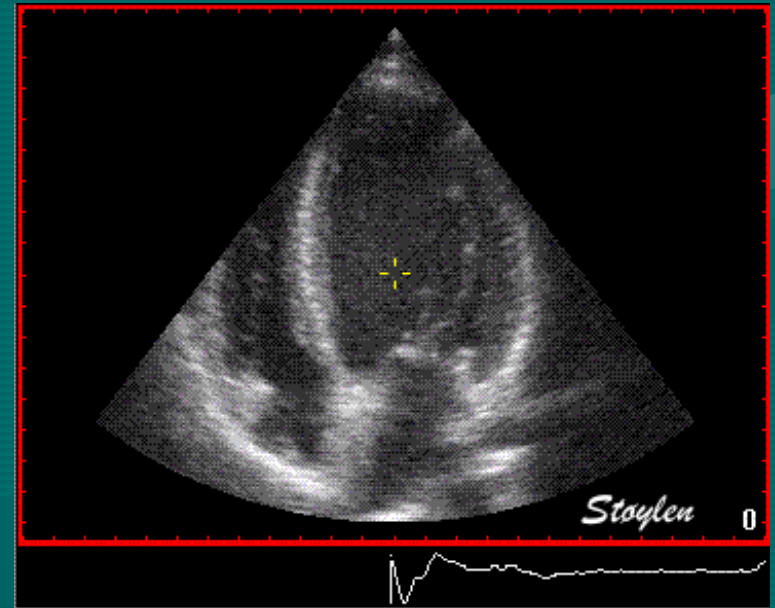
3-D Images



A 3-D ultrasound uses the same principle as a 2-D ultrasound. The difference is that the sound waves are emitted from all angles. The example above shows a 12 week fetal ultrasound images in the sagittal, axial and coronal planes that are used by the computer to generate the final 3D image in the lower right.

4-D Images

- A 4-D ultrasound uses the 3-D technology, but added in the time dimension. It displays a series of 3-D images over a period of time, thus giving it motion. These are generally short clips, and can show a heart beating or a baby yawning.



Advantages of Ultrasound Scans

- Can make images of soft tissues and to differentiate clearly between solids and fluid filled spaces.
- It makes diagnosis easy as it gives instant images so that the most useful can be selected by the operator.
- Allows for the structure of the organs to be detected as well as to determine how the organ is functioning, to some extent.
- There are no known side effects of this method, and the process does not cause any discomfort to the patient.
- The relatively small size of the scanners makes it possible to carry it anywhere.



Weaknesses of Ultrasound Scanning

- The basic ultrasound devices **cannot penetrate bones**; but ongoing programs are geared towards making it possible for bone imaging through ultrasound technology.
- **When a gas exists between the device and the target organ**, there is a lot of difficulty using ultrasound. This makes scanning of certain organs like the pancreas almost impossible.
- Ultrasound **cannot penetrate deep into the body**; this makes diagnosing organs that are deep in the body very difficult. The method depends highly on the operator who should be highly skilled and experienced in order to produce the quality images needed for the right diagnosis.
- There are **some concerns over the development of heat** during scanning
 - tissues or water will absorb the energy which **increases their temperature locally**.
 - the raised temperature may cause **formation of bubbles** (cavitation) when dissolved gases come out of solution due to the local heat.

Ultrasound (P4-June 2009) (1/3)

(a) Explain the main principles behind the use of ultrasound to obtain diagnostic information about internal body structures. [4]

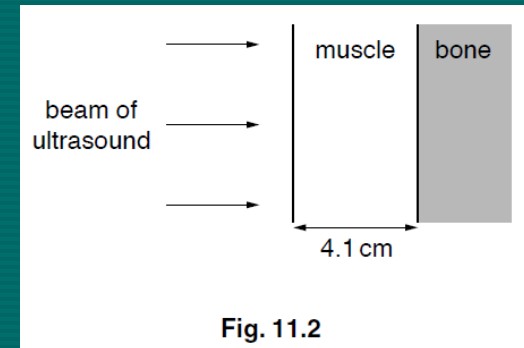
Solution:

- | | |
|---|-----|
| (a) <u>pulse</u> of ultrasound | (1) |
| reflected at boundaries / boundary | (1) |
| received / detected (at surface) by transducer | (1) |
| signal processed and displayed | (1) |
| time between transmission and receipt of pulse gives
(information about) depth of boundary | (1) |
| reflected intensity gives information as to nature of boundary | (1) |
| <i>(any four points, 1 each, max 4)</i> | |

Ultrasound (P4-June 2009) (2/3)

	acoustic impedance / $\text{kg m}^{-2} \text{s}^{-1}$	absorption coefficient / m^{-1}
muscle	1.7×10^6	23
bone	6.3×10^6	130

Fig. 11.1



- (b) Data for the acoustic impedances and absorption (attenuation) coefficients of muscle and bone are given in Fig. 11.1.

The intensity reflection coefficient is given by the expression

$$\frac{(Z_2 - Z_1)^2}{(Z_2 + Z_1)^2}.$$

The attenuation of ultrasound in muscle follows a similar relation to the attenuation of X-rays in matter. A parallel beam of ultrasound of intensity I enters the surface of a layer of muscle of thickness 4.1 cm as shown in Fig. 11.2.

The ultrasound is reflected at a muscle-bone boundary and returns to the surface of the muscle. Calculate

- (i) the intensity reflection coefficient at the muscle-bone boundary, [2]
- (ii) the fraction of the incident intensity that is transmitted from the surface of the muscle to the surface of the bone, [2]
- (iii) the intensity, in terms of I , that is received back at the surface of the muscle. [2]

Ultrasound (P4-June 2009) (3/3)

	acoustic impedance / $\text{kg m}^{-2} \text{s}^{-1}$	absorption coefficient / m^{-1}
muscle	1.7×10^6	23
bone	6.3×10^6	130

Fig. 11.1

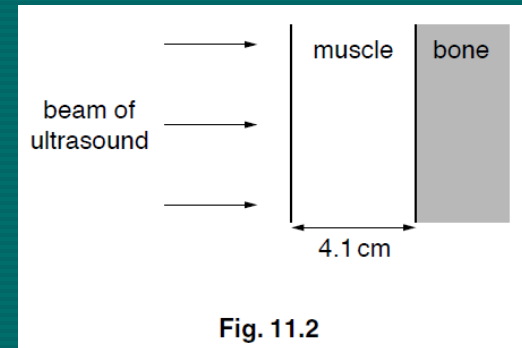


Fig. 11.2

Solution:

- (b) (i) coefficient = $(Z_2 - Z_1)^2 / (Z_2 + Z_1)^2$
 = $(6.3 - 1.7)^2 / (6.3 + 1.7)^2$ C1
 = 0.33 (unit quoted, then -1) A1 [2]
- (ii) fraction = $\exp(-\mu x)$ C1
 = $\exp(-23 \times 4.1 \times 10^{-2})$
 = 0.39 A1 [2]
- (iii) intensity = $0.33 \times 0.39^2 \times I$ C1
 = $0.050 I$ A1 [2]

(do not allow e.c.f. from (i) and (ii) if these answers are greater than 1)

Ultrasound (P4-Nov 2007)

- (a) State what is meant by *acoustic impedance*. [1]
- (b) Explain why acoustic impedance is important when considering reflection of ultrasound at the boundary between two media. [2]
- (c) Explain the principles behind the use of ultrasound to obtain diagnostic information about structures within the body. [5]

Solution:

- (a) product of density (of medium) and speed of sound (in medium)B1
- (b) difference in acoustic impedanceM1
determines fraction of incident intensity
that is reflected/amount of reflectionA1
- (c) pulse of ultrasound (directed into body)B1
reflected at boundary (between tissues)B1
(reflected pulse is) detected and processedB1
time for return of echo gives (information on) depthB1
amount of reflection gives information on tissue structuresB1

X-Ray

X-ray has long been used to take pictures of broken bones

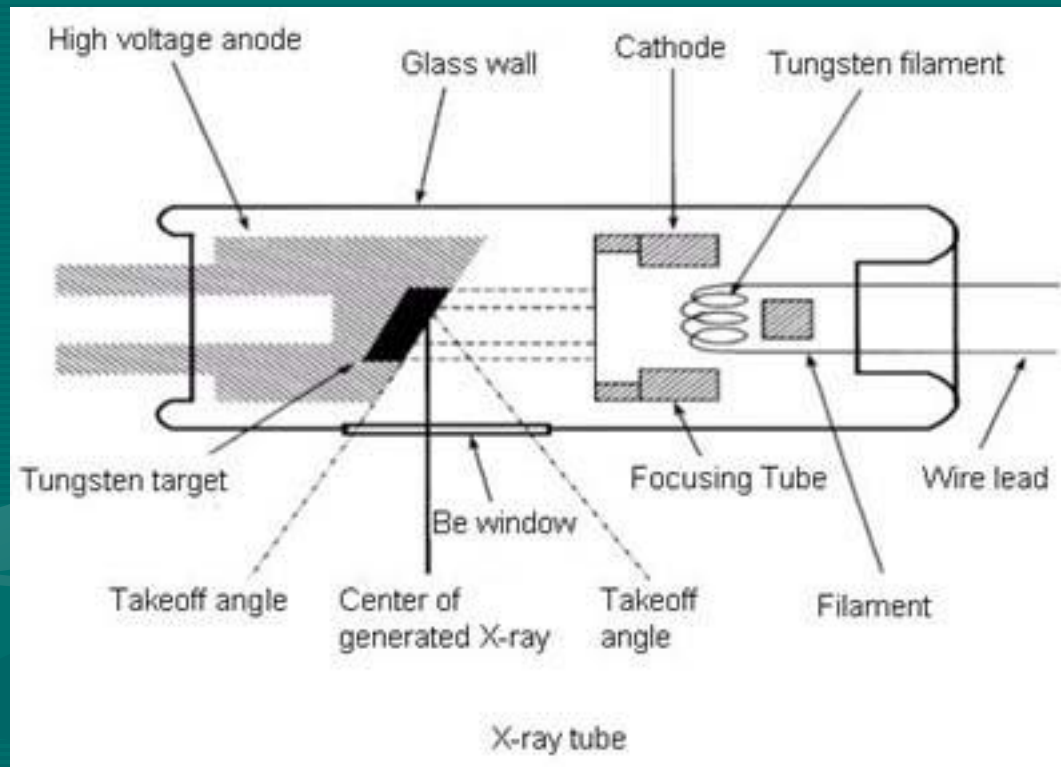


Production of X-Ray

Electrons emitted at the cathode is **accelerated** through the vacuum tube to hit the metal block anode.

On hitting the target 90% of the energy is converted to heat, 10% or less to **X-ray**

The anode has to be **cooled** by various methods.



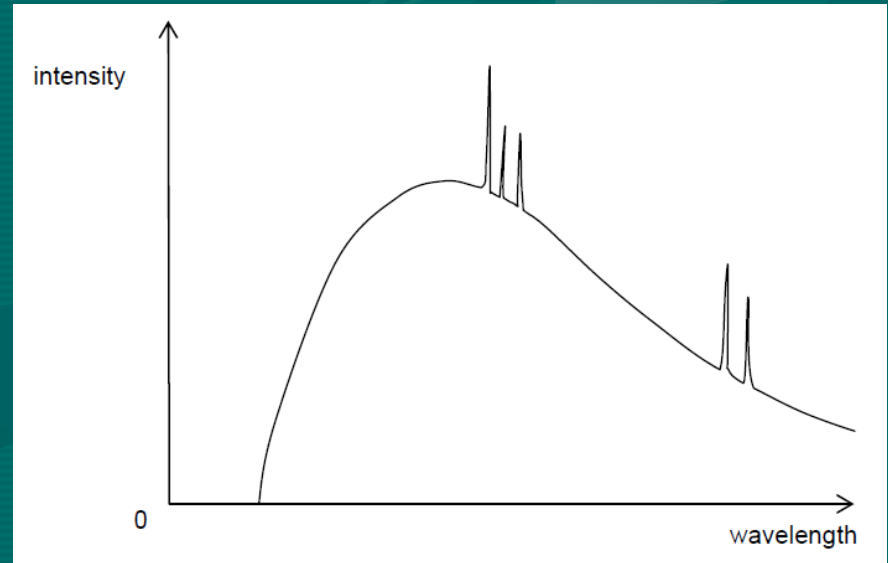
Thermionic Emission: The cathode is heated by electrical means and electrons are emitted

To produce X-ray, p.d. between anode and cathode must be **20 kV– 100 kV**

Production of X-Ray

X-rays are produced by two main mechanisms and come in two varieties.

- Bremsstrahlung X-rays
- Characteristic X-rays



The resultant spectrum has two components

Bremsstrahlung X-rays

Bremsstrahlung is a German word meaning “**braking radiation**” which describes the process of X-ray generation.

The high speed electron impacts on the target and at the atomic level approaches the nucleus.

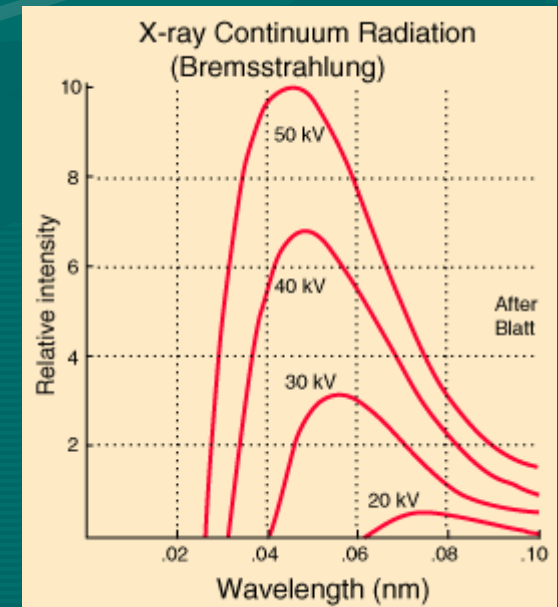
There is no actual collision between electron and nucleus because the electron interacts with the Coulombic nuclear forces and its **vector quantities of direction and velocity** are changed.

The **change in energy is radiated as electromagnetic radiation**. The large amount of energy means a short wavelength within the X-ray band.

As the electron is not destroyed, it can undergo multiple interactions, and even initial interactions will vary from minor to major energy changes depending on the actual angle and proximity of attack, and the point of 'impact' on the nucleus.

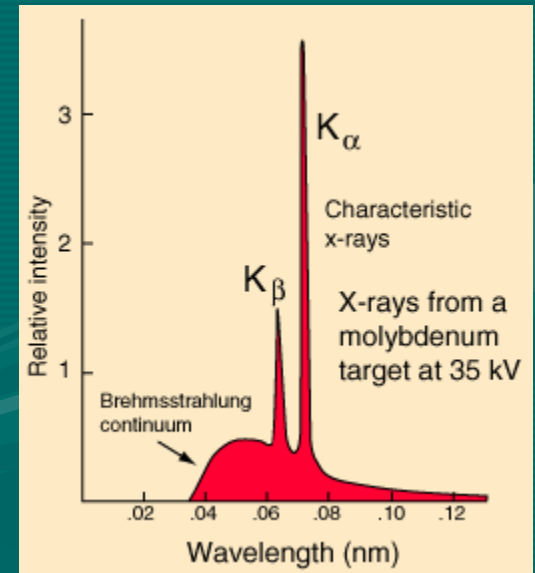
As a result, bremsstrahlung radiation will have **continuous spectrum** where the maximum energy relates to the entire KE of the electron.

maximum kinetic energy of an electron = $eV = hc / \lambda$

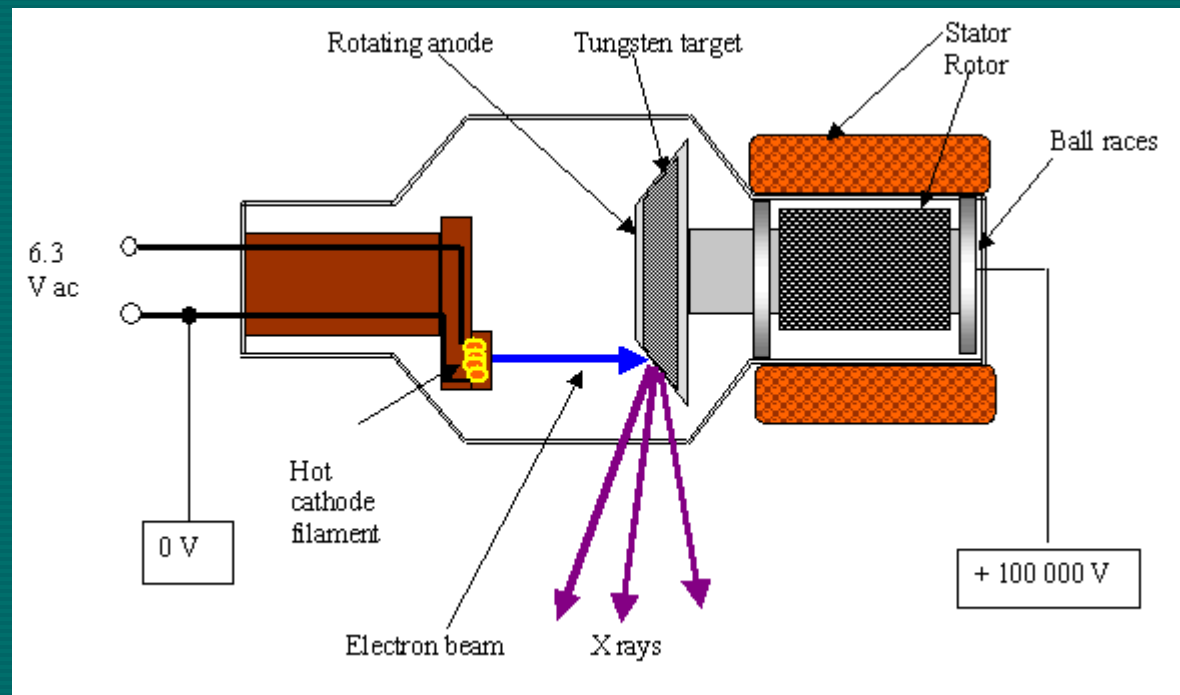


Characteristic X-rays

- Some of the **bombarding electrons** will **collide** with the **orbiting electrons**. Sufficient energy in such collisions can result in the **ejection of an orbiting electron**. 'Sufficient energy' means enough to overcome the bonding energy of the orbiting electron.
- The impacting electron will move off with reduced energy, and the ejected electron will move off in a different direction and speed with the remaining energy,
- There is **an empty position** in one of the shells. The remaining orbiting electrons will '**pack down**' to fill the hole, and when changing orbits will **lose energy** and **emit this as radiation**.
- The **orbiting levels are fixed** as a physical property fixing the elemental identity of an atom, and so the energy emission will be **characteristic of that atom**.
- The energy will be mono-energetic and so **appear as a spike** rather than a continuous spectrum. Electrons ejected come from the $n = 1, 2$ and 3 orbits. The atom becomes an ion as it has lost an ejected electron.
- All atoms will produce characteristic radiation but not all are visible in the X-ray portion of the electromagnetic spectrum. **Tungsten** and **Molybdenum** have theirs in the X-ray region.



Cooling of the Anode



The anode is either water-cooled or is made to spin rapidly so that the target area is increased

Intensity of the X-ray beam

- The intensity of the X-ray beam is determined by the rate of arrival of electrons at the metal target, that is, the *tube current*.
- This tube current is controlled by the heater current of the cathode.
- The greater the heater current, the hotter the filament and hence the greater the rate of emission of thermo-electrons.

Hardness of the X-ray beam

- The **hardness** of an X-ray beam refers to its **penetration power**.
- The hardness is **controlled by the accelerating voltage between the cathode and the anode**.
- More penetrating X-rays have higher photon energies and thus a larger accelerating potential is required.
- Referring to the spectrum of X-rays produced, it can be seen that longer wavelength X-rays (**'softer' X-rays**) are also produced.
- These X-ray photons are of such **low energy** that they would not be able to pass through the patient.
- They would contribute to the total radiation dose without any useful purpose.
- Consequently, an **aluminium filter** is frequently fitted across the window of the X-ray tube to **absorb the 'soft' X-ray photons**.

Example

The accelerating potential difference between the cathode and the anode of an X-ray tube is 30 kV. Given that the Planck constant is $6.6 \times 10^{-34} \text{ J s}$, the charge on the electron is $1.6 \times 10^{-19} \text{ C}$ and the speed of light in free space is $3.0 \times 10^8 \text{ m s}^{-1}$, calculate the minimum wavelength of photons in the X-ray beam.

Solution:

For the minimum wavelength,
energy gained by electron = energy of photon

$$eV = \frac{hc}{\lambda_0}$$

$$1.6 \times 10^{-19} \times 30 \times 10^3 = \frac{(6.6 \times 10^{-34} \times 3.0 \times 10^8)}{\lambda_0}$$

$$\lambda_0 = 4.1 \times 10^{-11} \text{ m}$$

X-ray Imaging

- X-ray radiation affects photographic plates
- X-ray beams are used to obtain 'shadow' pictures of the inside of the body to assist in the diagnosis or treatment of illness.
- If a picture is required of bones, this is relatively simple since the absorption by bone of X-ray photons is considerably greater than the absorption by surrounding muscles and tissues.
- X-ray pictures of other parts of the body may be obtained if there is sufficient difference between the absorption properties of the organ under review and the surrounding tissues.



Quality of the Image

- The **quality** of the shadow picture (the image) produced on the photographic plate depends on its **sharpness** and **contrast**.
- **Sharpness** is concerned with the **ease with which the edges of structures can be determined**. A sharp image implies that the edges of organs are clearly defined.
- An image has **good contrast** if there is a marked **difference in the degree of blackening** of the image between one organ and another.

To Obtain Sharp Images

The X-ray tube is designed to generate a beam of X-rays with minimum width. Factors in the design of the X-ray apparatus that may affect sharpness include:

- the area of the target anode, as illustrated in Fig. 2.3,

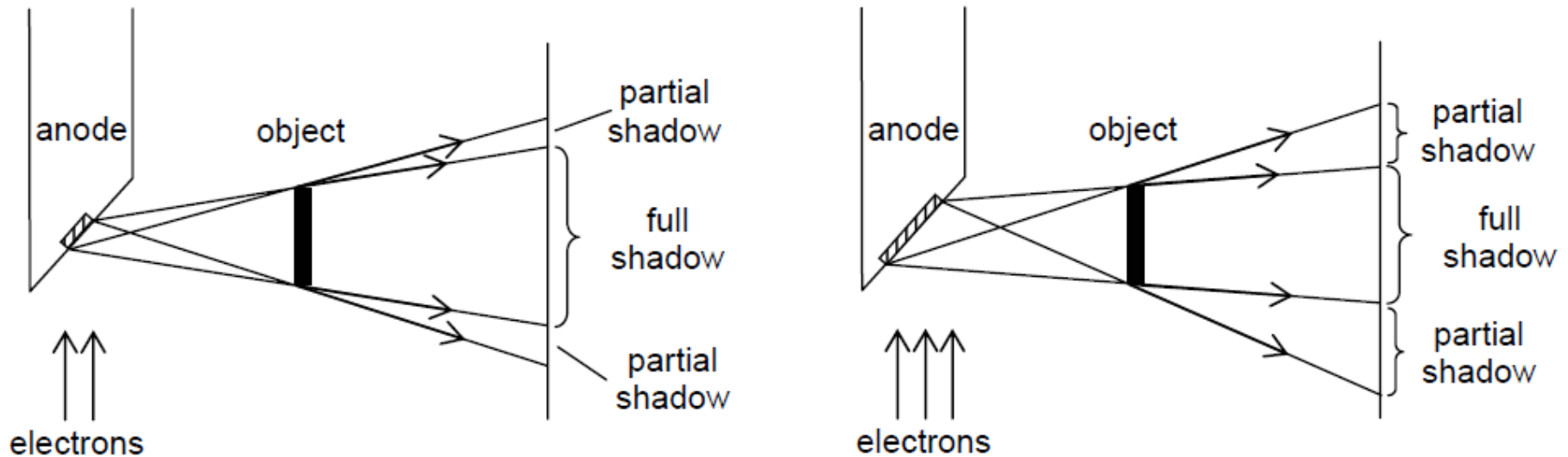


Fig. 2.3

To Obtain Sharp Image

- the size of the aperture, produced by overlapping metal plates, through which the X-ray beam passes after leaving the tube (see Fig. 2.4),

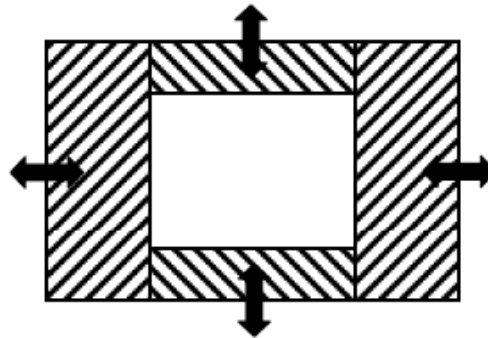


Fig. 2.4

To Obtain Sharp Image

- the use of a lead grid in front of the photographic film to absorb scattered X-ray photons, as illustrated in Fig. 2.5.

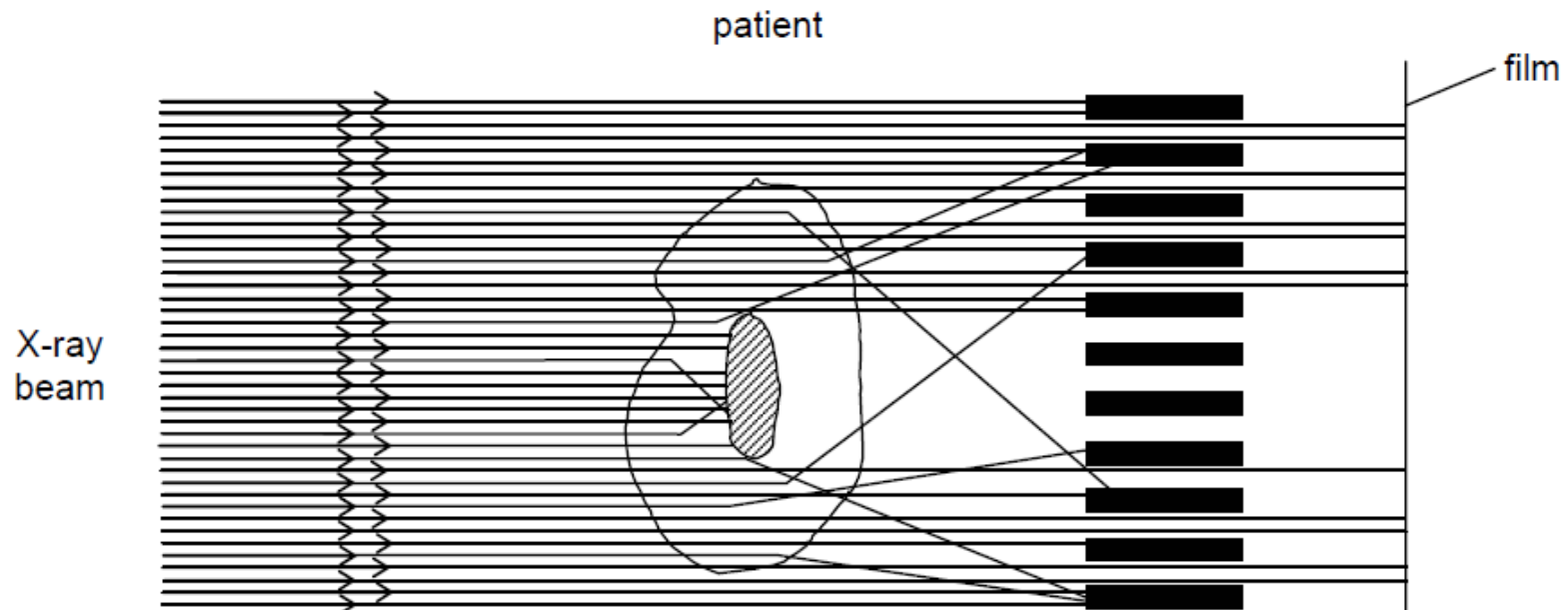


Fig. 2.5

To Obtain Good Contrast

- Use a ‘contrast medium’. For example, the stomach may be examined by giving the patient a drink containing barium sulphate. Similarly, to outline blood vessels, a contrast medium that absorbs strongly the X-radiation would be injected into the bloodstream.
- The contrast of the image produced on the photographic film is affected by
 - exposure time,
 - X-ray penetration and
 - scattering of the X-ray beam within the patient’s body.
- Contrast may be improved by backing the photographic film with a fluorescent material.

Attenuation of X-ray

- Attenuation refers to the reduction of intensity.
- The intensity of the X-rays is reduced as it travels through a medium.

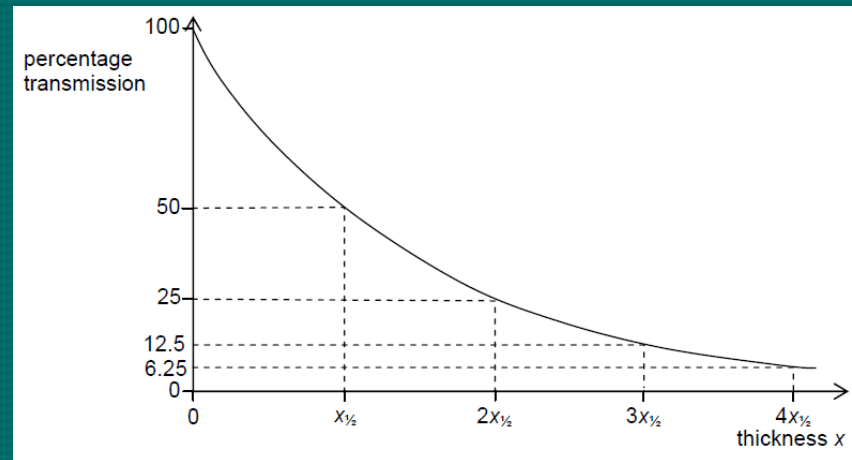
$$I = I_0 e^{-\mu x}$$

μ is the *linear absorption coefficient* or *linear attenuation coefficient* of the medium.

The unit of μ is mm^{-1} or cm^{-1} or m^{-1} .

x is the thickness of the medium passed through

Half-value Thickness (HVT)



- The *half-value thickness* $x_{1/2}$ or *HVT* is the thickness of the medium required to reduce the transmitted intensity to one half of its initial value.
- It is a constant and is related to the linear absorption coefficient μ by the expression
$$x_{1/2} \times \mu = \ln 2.$$
- In practice, $x_{1/2}$ does not have a precise value as it is constant only when the beam has photons of one energy only.

Example

The linear absorption coefficient of copper is 0.693 mm^{-1} . Calculate:

- (a) the thickness of copper required to reduce the incident intensity by 50% (the half-value thickness),
- (b) the fraction of the incident intensity of a parallel beam that is transmitted through a copper plate of thickness 1.2 cm.

Solution:

$$\text{(a)} \quad I = I_0 e^{-\mu x} \text{ or } I = I_0 \exp(-\mu x_{\frac{1}{2}})$$

$$I / I_0 = 0.50 = \exp(-0.693 x_{\frac{1}{2}})$$

$$\ln 0.50 = -0.693 x_{\frac{1}{2}}$$

$$x_{\frac{1}{2}} = \mathbf{1.0 \text{ mm}}$$

$$\text{(b)} \quad I / I_0 = \exp(-0.693 \times 12)$$

$$I / I_0 = \mathbf{2.4 \times 10^{-4}}$$

Homework

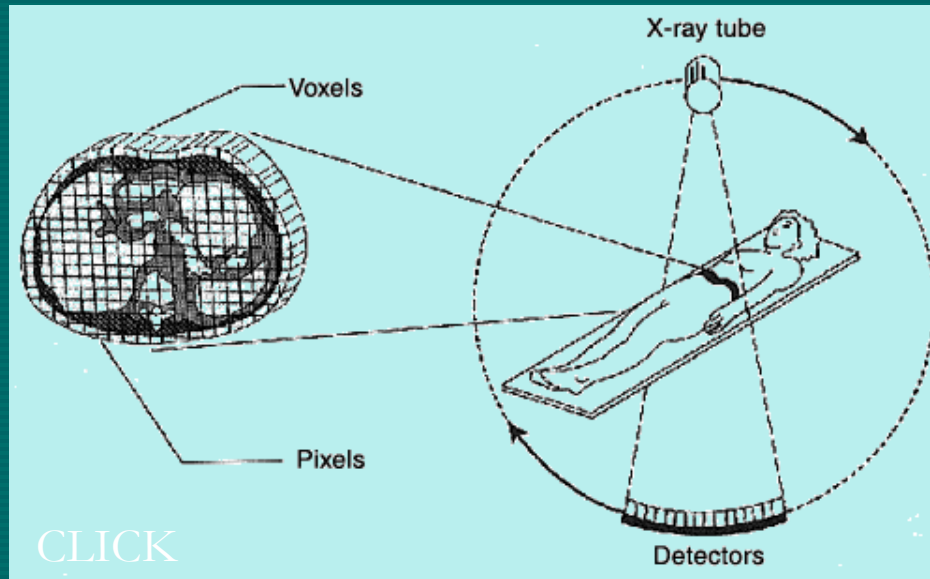
Compare the imaging process of X-ray with that of MRI, CT and ultrasound.

List its advantages and disadvantages compared to each of them.

COMPUTED TOMOGRAPHY

CT SCAN

What is a CT Scan?



- ❖ **CT** scanning or **computed tomography**, involves an **X-ray tube** that **turns 360 degrees** around the patient. (Also known as CAT)
- ❖ In this way, information is acquired from different angles in a plane.
- ❖ This information is reconstructed by computer technique into a **slice of cross-sectional image of the body**.
- ❖ Images of **successive slices can be combined to give a three-dimensional image**. The three-dimensional image can be rotated and viewed from any angle.

Use of CT Scans



CT Image of the
Abdomen



CT Image of the
Thorax - Aorta

- CT is fast, patient-friendly and has the unique ability to image a combination of soft tissue, bone, and blood vessels.
- Physician can selectively "window" the digital CT images on the computer monitor to look at the soft tissue, then the bone and then the blood vessels, as needed.
- For example, CT is used extensively for diagnosing problems of the inner ears and sinuses because the anatomy of the inner ear and sinuses is made up of delicate soft tissue structure and very fine bones.
- It is also the preferred method for diagnosing lung, liver and pancreas cancer.

A CT Scan Demo



Advances of CT Scan

- Original CT scanners (1974 to 1987) would spin 360° in one direction and make an image (or slice), then spin 360° in the other direction to make a second slice. Between each slice, the machine would stop completely and reverse directions while the patient table was moved forward by an increment equal to the thickness of a slice.
- In the mid-1980s, an innovation called the "power slip ring" allowed scanners to rotate continuously. This development led to a new type of CT called "spiral" or "helical" scanning.
- The spiral or helical scanning dramatically increased its speed and effectiveness.

The Helical Scanning

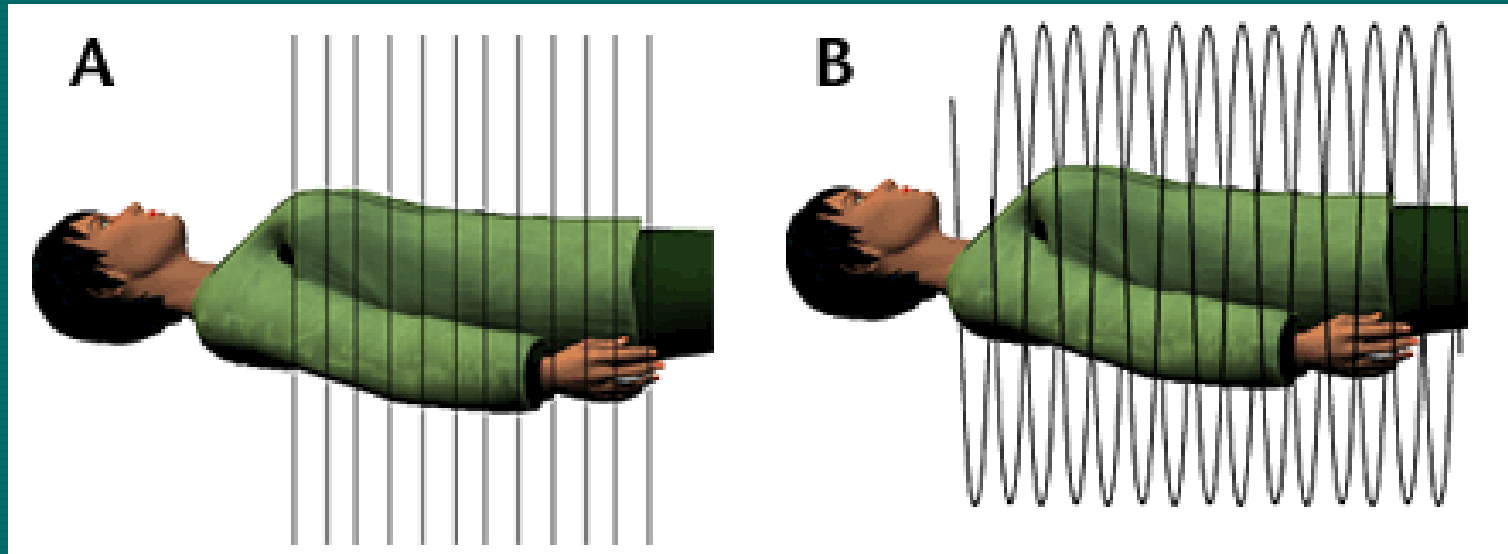


Figure A: Conventional CT Scanning

Conventional CT scans take pictures of slices of the body (like slices of bread). These slices are a few millimeters apart.

Figure B: Helical CT Scanning

The helical CT scan takes continuous pictures of the body in a rapid spiral motion, so that there are no gaps in the pictures collected.

Structure of a CT Scanner

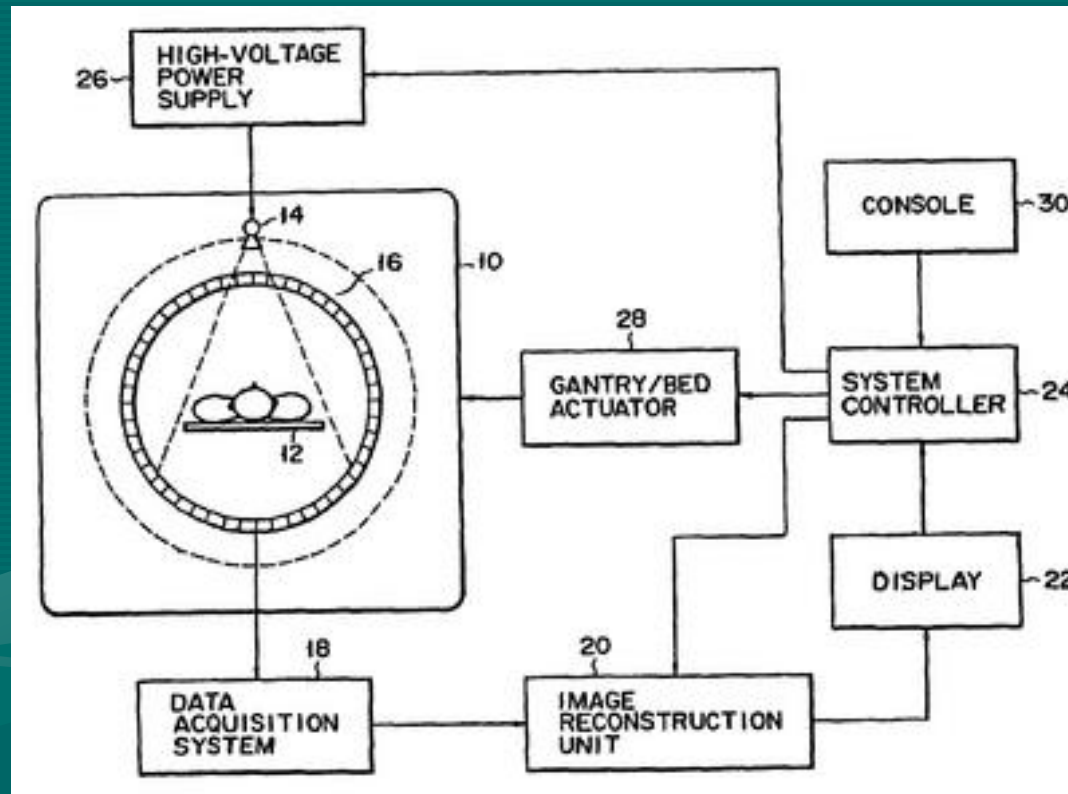


The Outside View of a
Modern CT System

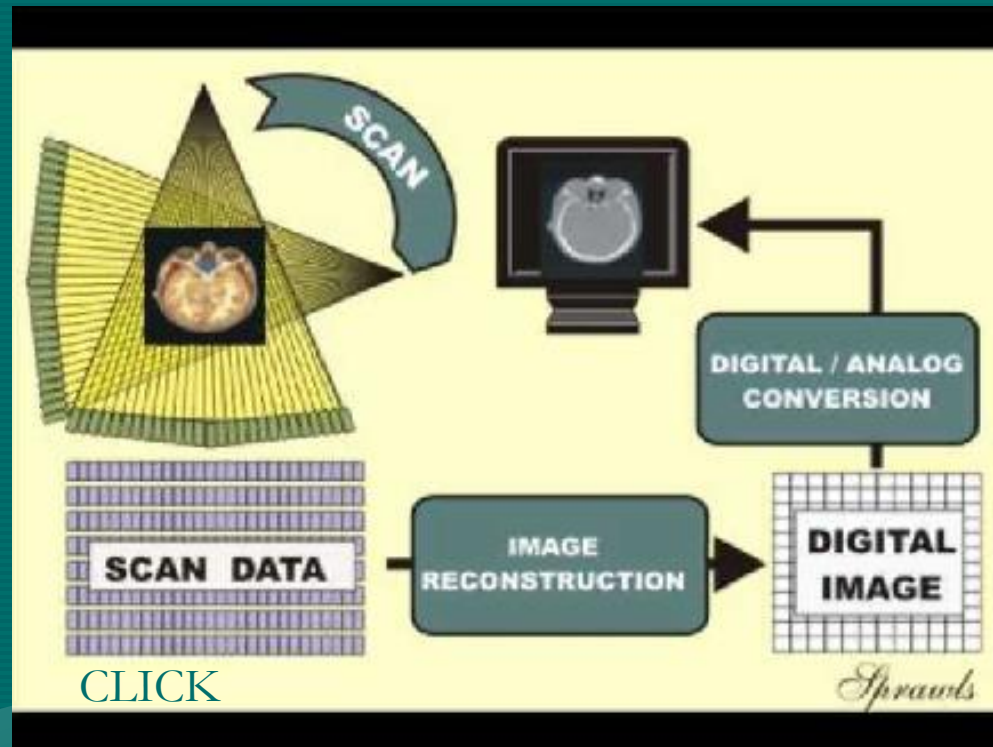


The Inside View of a
Modern CT System

Block Diagram



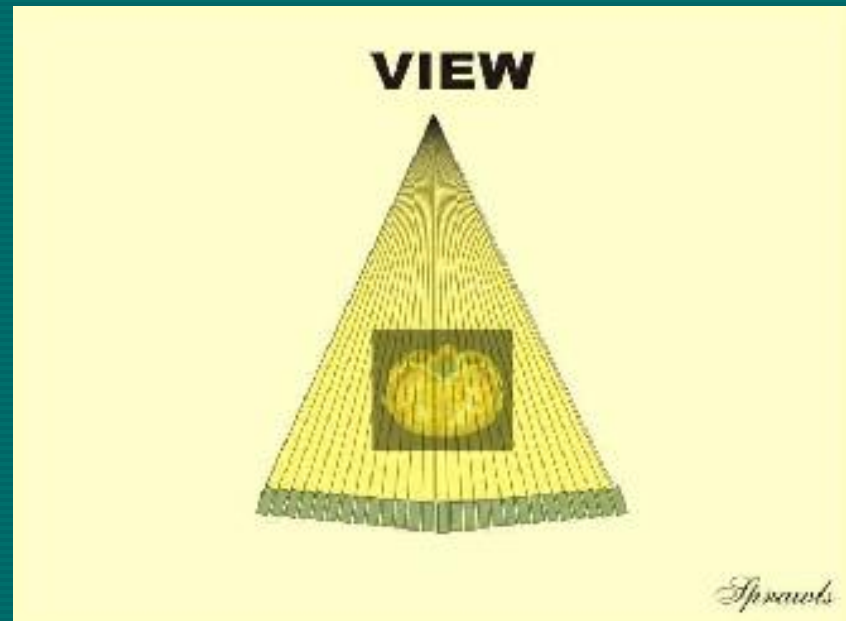
CT Image Formation



The formation of a CT image is a distinct three phase process.

- ❖ The **scanning phase** produces data, but not an image.
- ❖ The **reconstruction phase** processes the acquired data and forms a digital image.
- ❖ The visible and displayed analog image (shades of grey) is produced by the

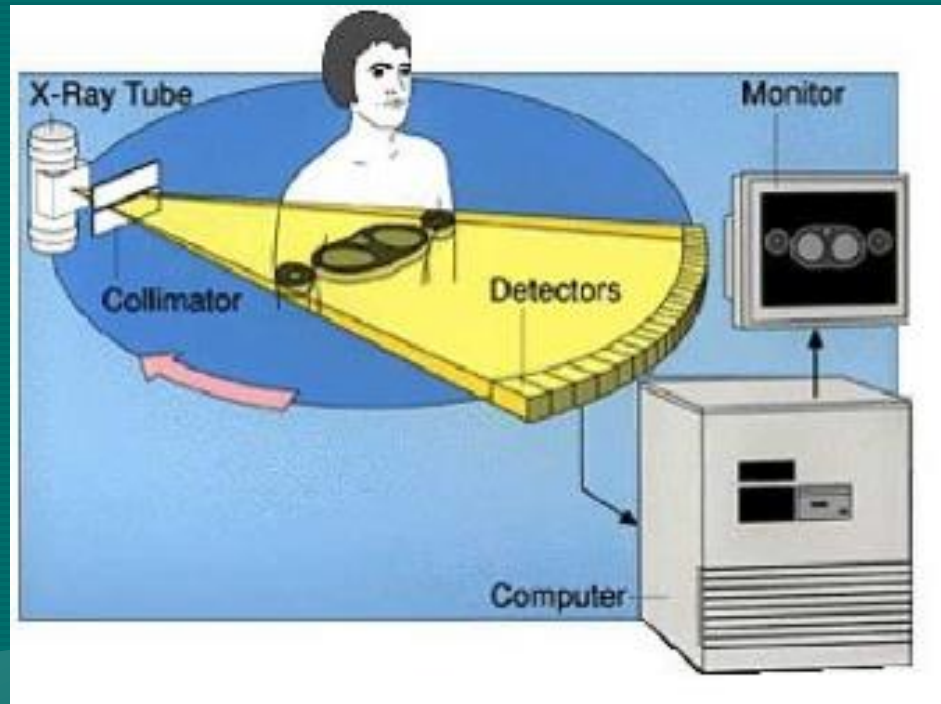
A CT X-ray Beam View



The projection of the **fan-shaped x-ray beam** from one specific x-ray tube focal spot position produces one view.

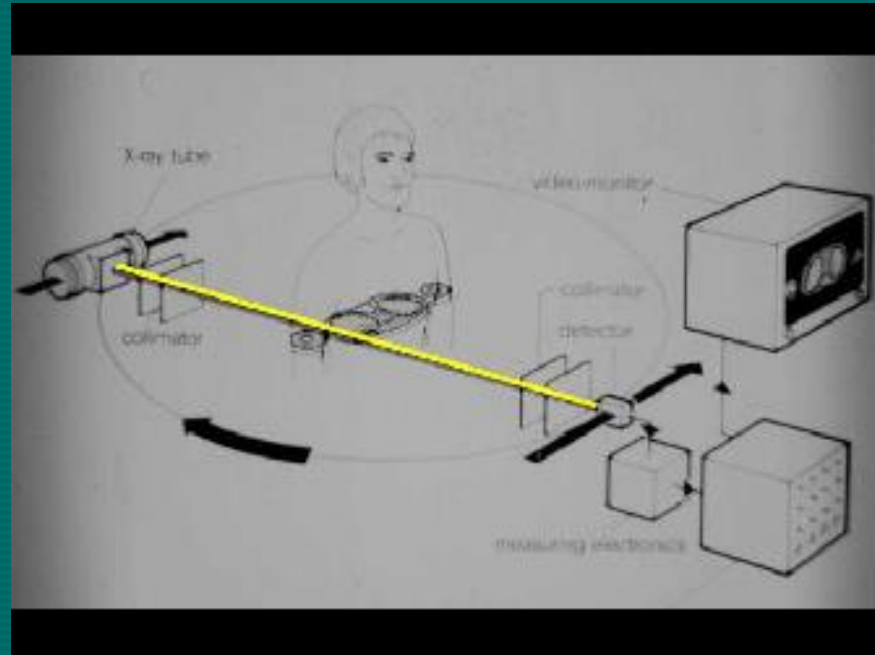
Many views projected from around the patient's body are required in order to acquire the necessary data **to reconstruct an image**.

The CT Imaging Process Using Views



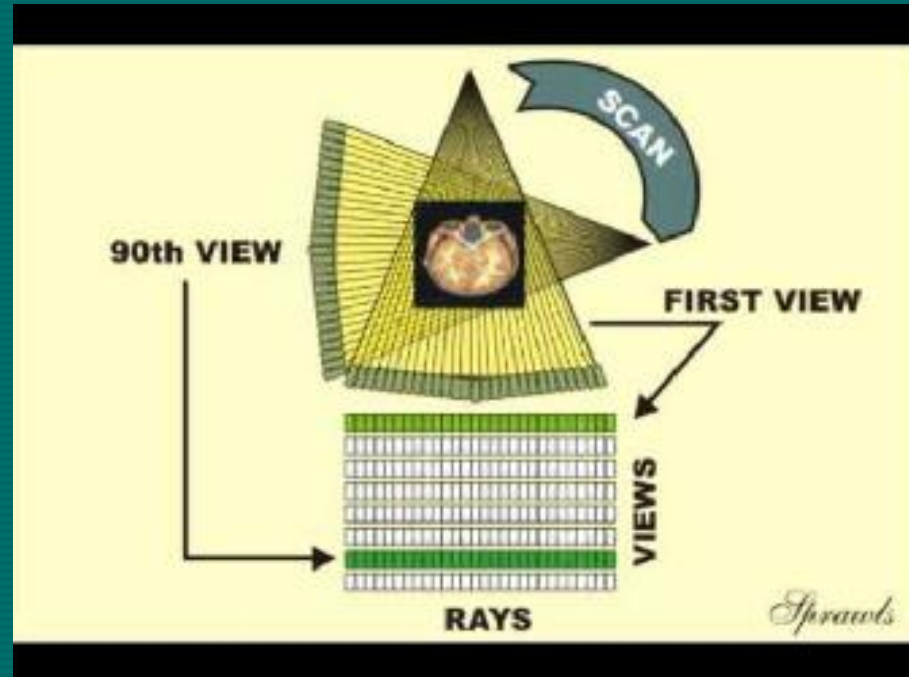
As the x-ray beam is scanned around the body, forming **many views**, the data recorded by the detectors are **stored in a computer memory** for later image reconstruction.

A Ray



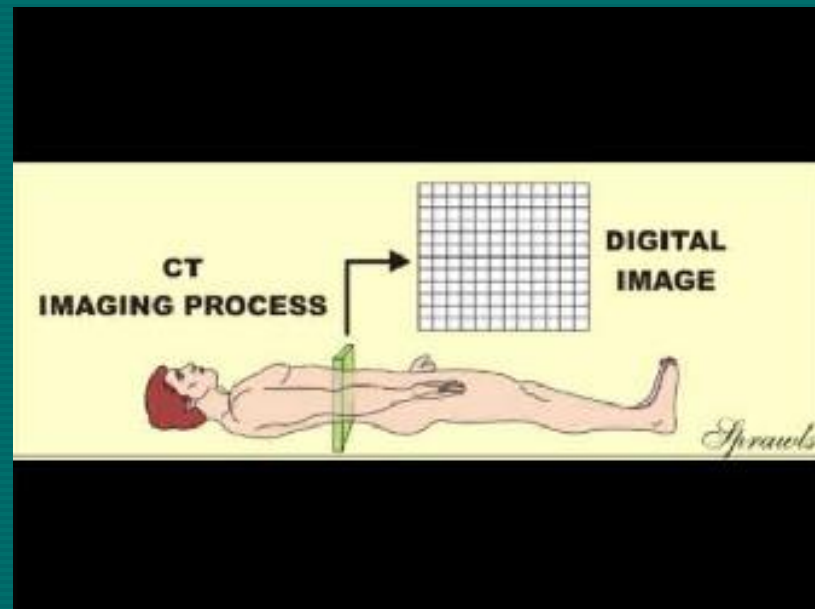
- ❖ A ray is the pathway of a portion of the x-ray beam from one specific focal-spot position to a specific detector position.
- ❖ As the ray passes through the body, it measures the total x-ray attenuation (or penetration) along it's path. This is the data recorded by the detector.
- ❖ A view, as seen previously, is made-up of many individual rays.

A Complete Scan



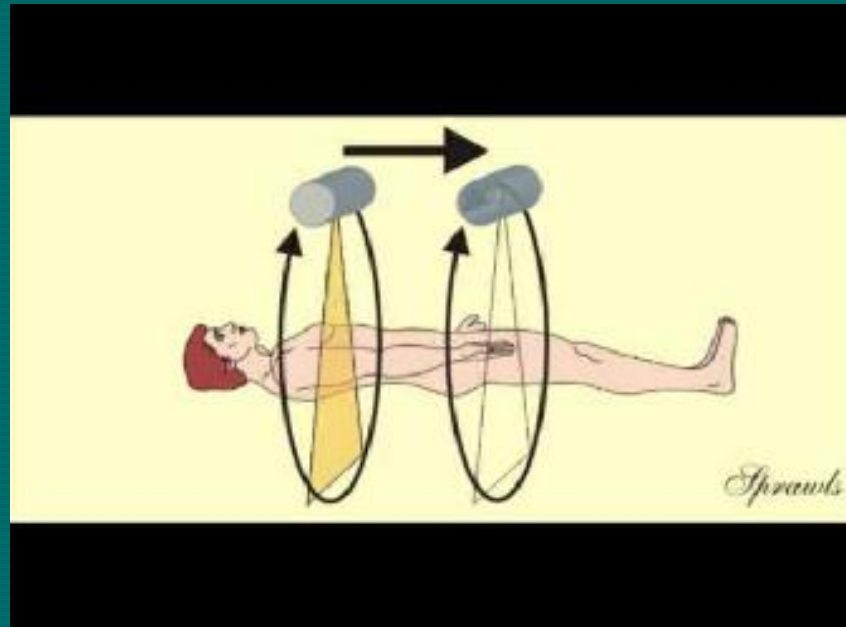
- ❖ A complete scan is formed by rotating the x-ray tube completely around the body and projecting many views.
- ❖ Each view produces one "profile" or line of data as shown here.
- ❖ The complete scan produces a complete data set that contains sufficient information for the reconstruction of an image.
- ❖ In principle, one scan produces data for one slice image. However, with spiral/helical scanning, there is not always a one-to-one relationship between the number of scans around the body and the number of slice images produced.

The CT Image



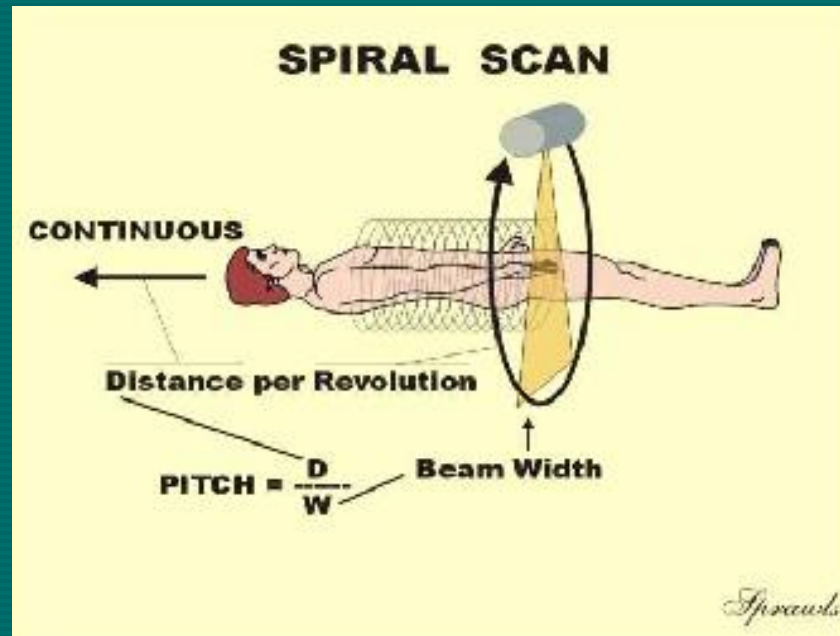
- ❖ The principle objective of CT imaging is to produce a digital image (a matrix of pixels) for a specific slice of tissue.
- ❖ During the image reconstruction process, the slice of tissue is divided into a matrix of voxels (volume elements).
- ❖ As we will see later, a CT number is calculated and displayed in each pixel of the image. The value of the CT number is calculated from the x-ray attenuation properties of the corresponding tissue voxel.

X-ray Tube Motions



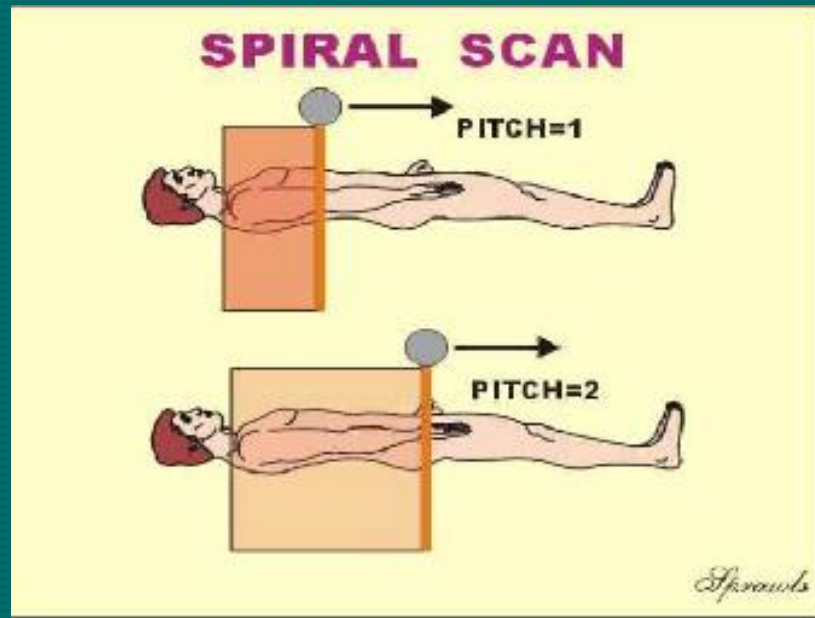
- ❖ There are **two distinct motions** of the x-ray beam relative to the patient's body during CT imaging.
- ❖ One motion is the **scanning of the beam around the body** as we have just seen.
- ❖ The other motion is the **movement of the beam along the length of the body**. This is achieved by moving the body through the beam as it is rotating around.

Spiral/Helical Scanning



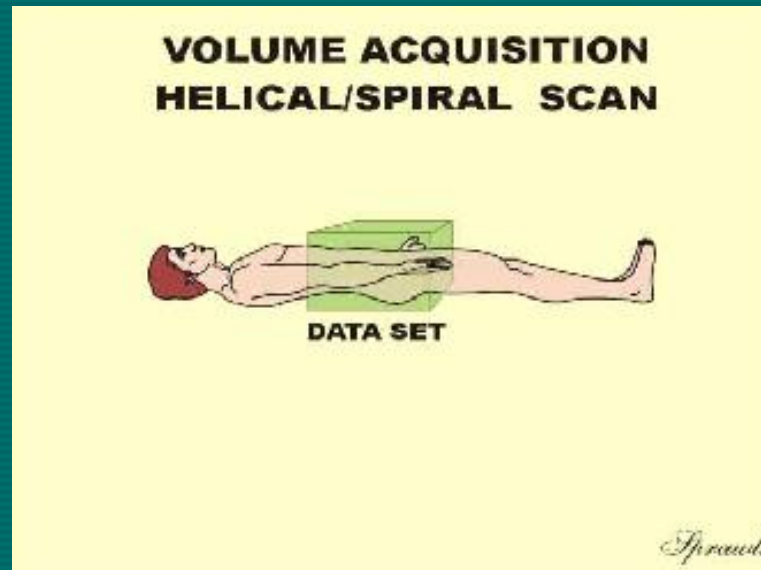
- ❖ **Spiral or helical scanning** is a more recently developed mode and is used for many procedures.
- ❖ The patient's **body is moved continuously** as the x-ray beam is scanned around the body.
- ❖ This motion is controlled by the operator selected value of the **pitch factor**.
- ❖ As illustrated, the pitch value is **the distance the body is moved** during one beam rotation, **expressed as multiples of the x-ray beam width** or thickness.
- ❖ If the body is moved 10 mm during one rotation, and the beam width is 5 mm, the pitch will have a value of 2.

Changing the Pitch



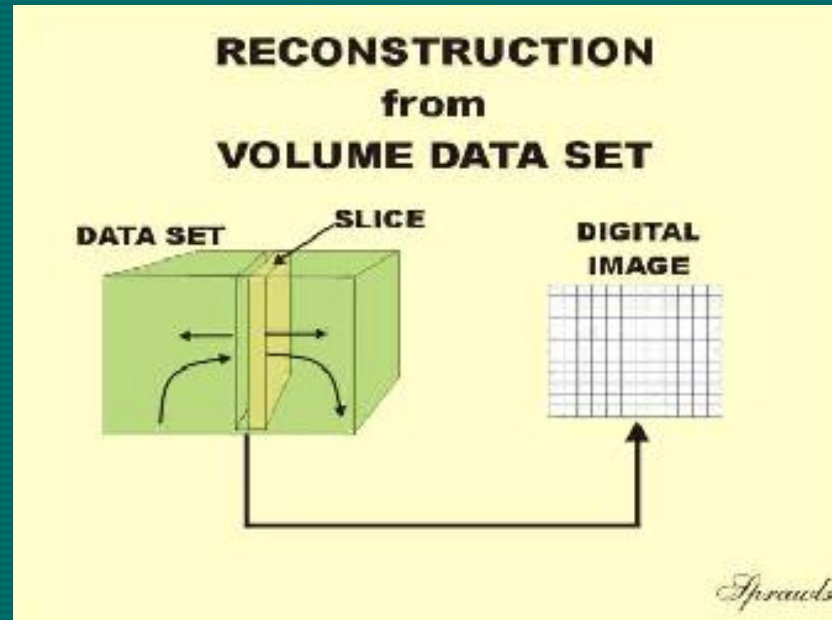
- ❖ As we see here, when the pitch is increased, the x-ray beam appears to move faster along the patient's body.
- ❖ During the same time (as illustrated), the x-ray beam will be spread over more of the body **when the pitch is increased**. This has three major effects.
 - **Scan time will be less** to cover a specific body volume.
 - The radiation is less concentrated so **dose is reduced**.
 - There will not be as much "detail" in the data and **image quality might be reduced**.

Volume Data Sets



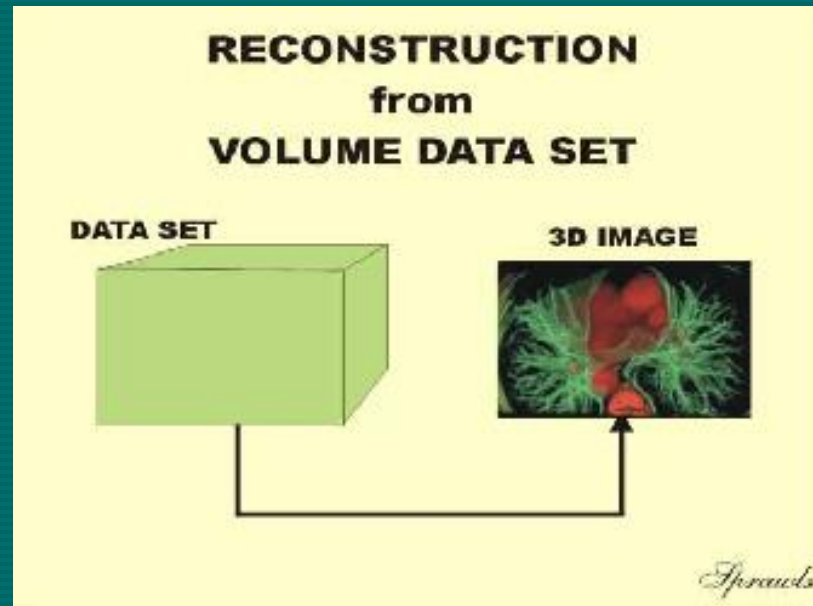
- ❖ A major advantage of spiral/helical scanning is that it produces a continuous data set extending over some volume of the patient's body.
- ❖ The data set is not broken up into slices as with the scan/step slice acquisition method.
- ❖ As we will soon see, the **volume data set can be sliced many ways later during the image reconstruction phase.**

Reconstruction from Volume Data Sets



- ❖ A **major advantage** of spiral scanning is that the **thickness, position, and orientation of image slices can be adjusted** during the reconstruction phase.
- ❖ Images of overlapping slices can be created.
- ❖ The reconstruction can be repeated to produce images with different spatial characteristics.

3-D Image Reconstruction



A volume data set can be used to reconstruct 3-D images.

A general requirement for good-quality 3-D images is that the data set have "**good detail**" in the long patient axis direction. This is achieved by scanning with **thin beams** and relatively **low pitch values**.

Example 1: View 1

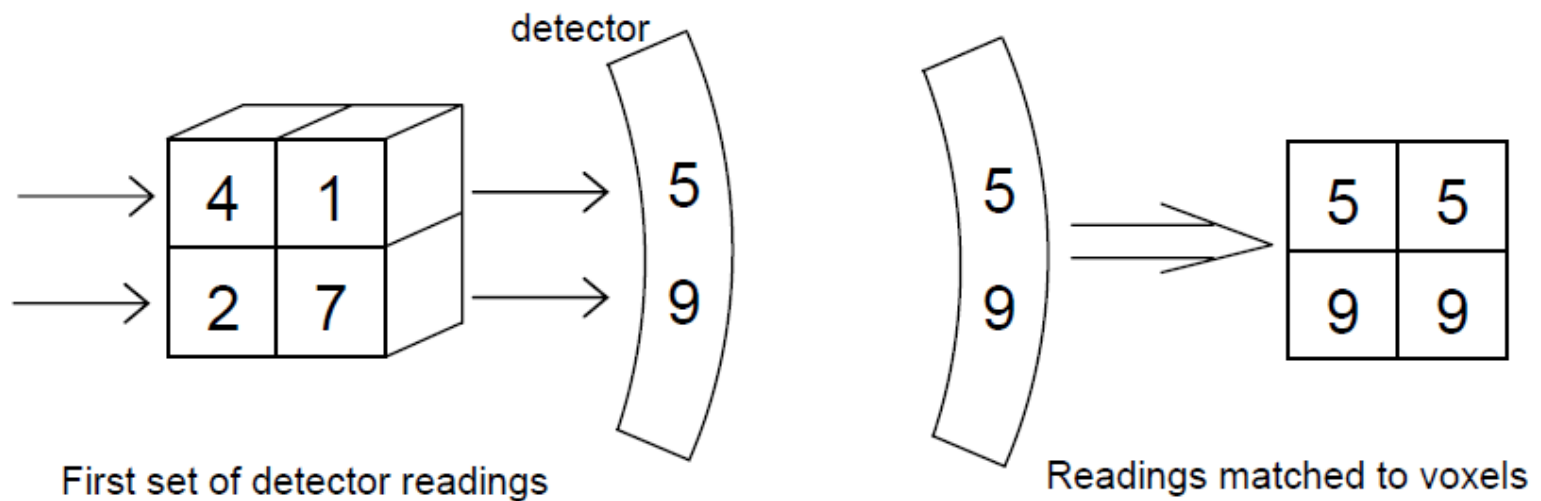


Fig. 2.9

Example 1: View 2 (45° clockwise)

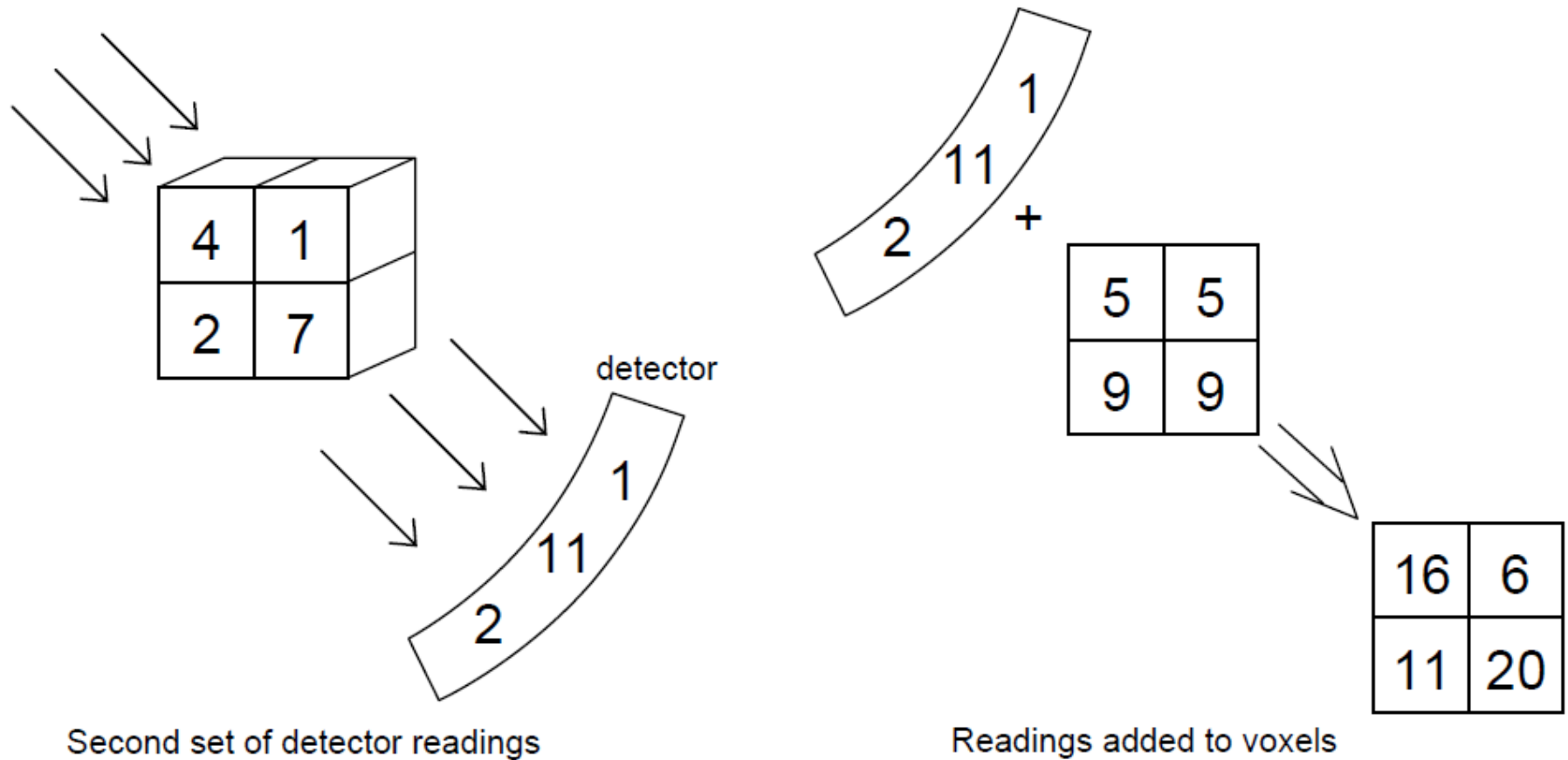
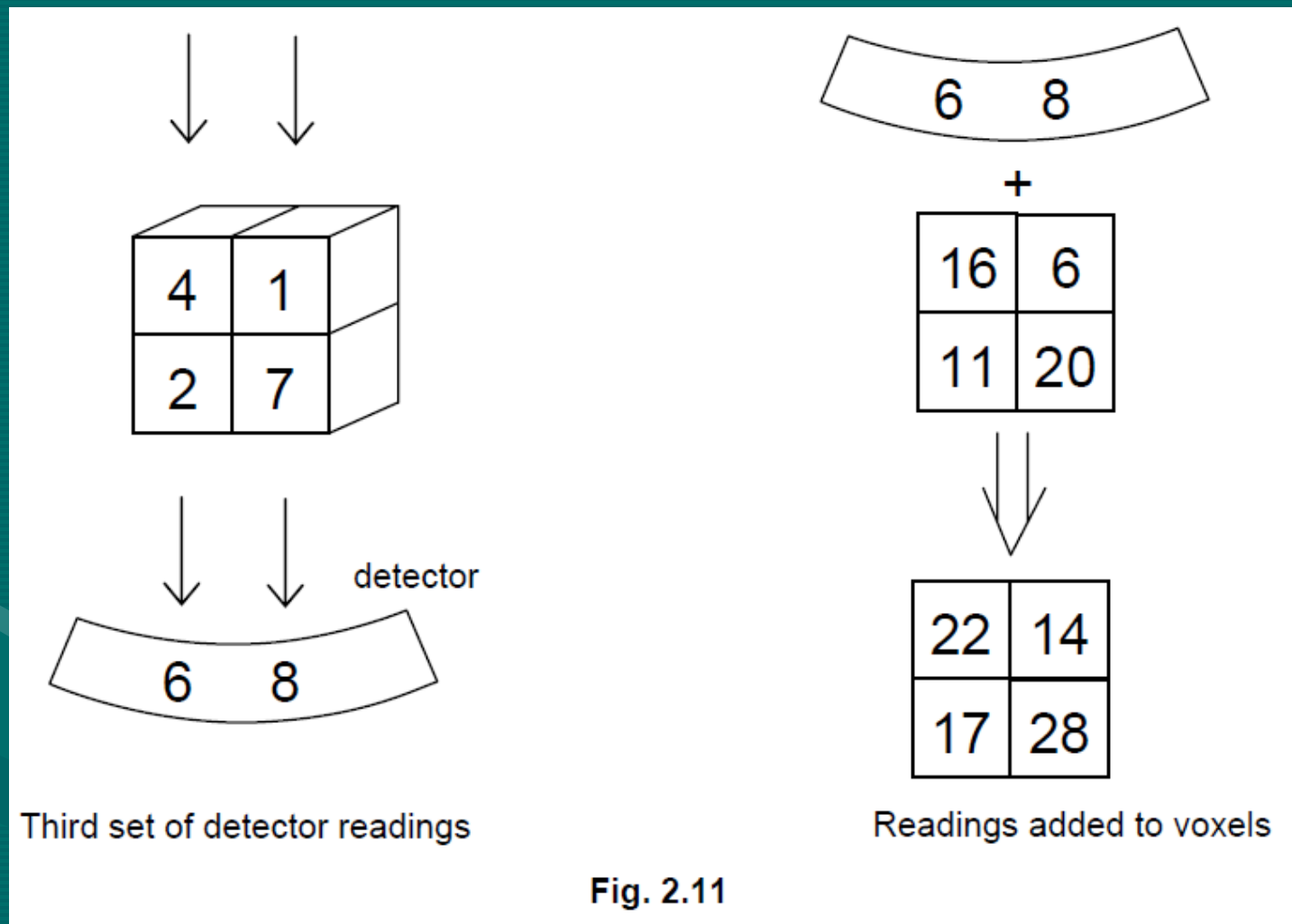
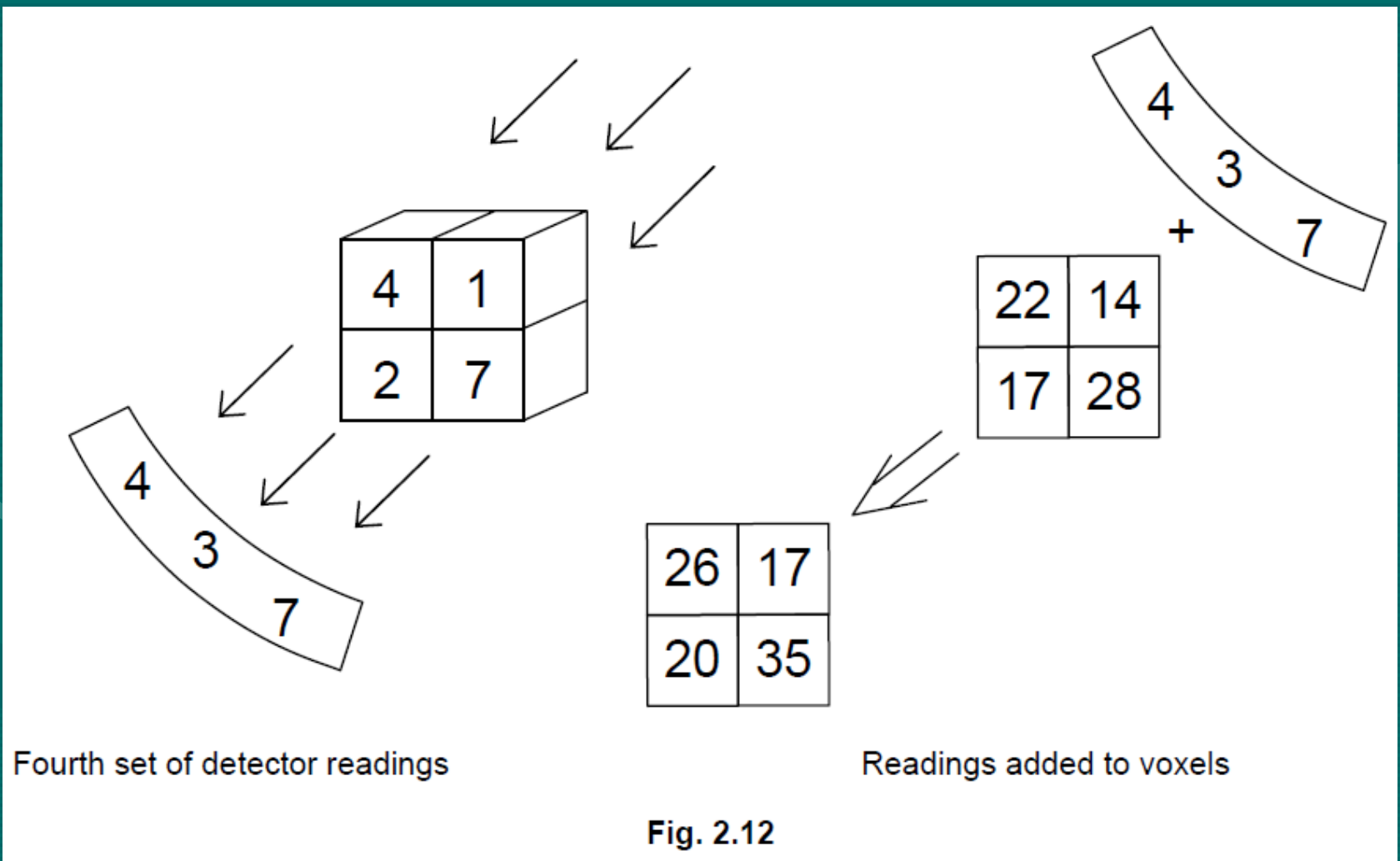


Fig. 2.10

Example 1: View 3 (90° clockwise)



Example 1: View 4 (135° clockwise)



Obtaining the Original Pattern

In order to obtain the original pattern of pixels, two operations must be performed.

1. The 'background' intensity must be removed. The 'background' intensity is the total of each set of detector readings. In this case, 14 is deducted from each pixel.
2. After deduction of the 'background', the result must be divided by three to allow for the duplication of the views of the section.

These processes are illustrated in Fig. 2.14.

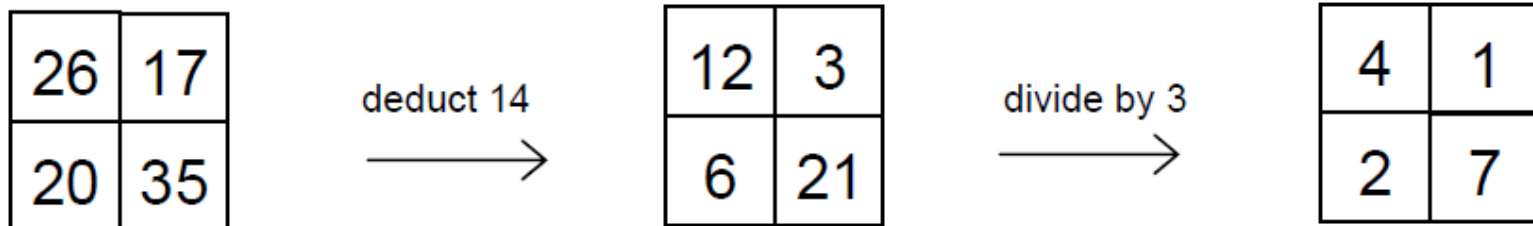
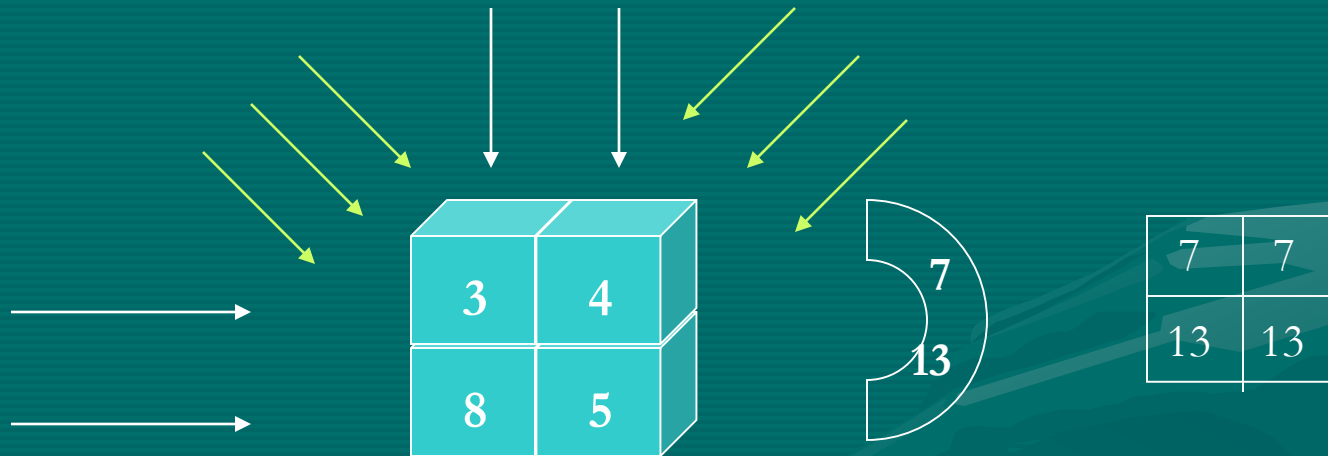


Fig. 2.14

Example 2



Try this example like the previous, you should get back the same pixel numbers.

CT Scan (P4-Nov 2008)

- (a) Distinguish between the images produced by CT scanning and X-ray imaging. [3]
- (b) By reference to the principles of CT scanning, suggest why CT scanning could not be developed before powerful computers were available. [5]

Solution:

- | | |
|--|-----|
| (a) CT image: (thin) slice (through structure) | B1 |
| any further detail e.g. built up from many 'slices' / 3-D image | B1 |
| X-ray image: 'shadow' image (of whole structure) / 2-D image | B1 |
| | |
| (b) X-ray image <u>of slice</u> taken from many different angles | (1) |
| these images are combined (and processed) | (1) |
| repeated for many different slices | (1) |
| to build up a 3-D image | (1) |
| 3-D image can be rotated | (1) |
| computer required to store and process huge quantity of data | (1) |
| (any five, 1 each to max 5) | B5 |

X-Ray (P42-Nov 2009) (1/2)

(a) A typical spectrum of the X-ray radiation produced by electron bombardment of a metal target is illustrated in Fig. 10.1.

Explain why

(i) a continuous spectrum of wavelengths is produced, [3]

(ii) the spectrum has a sharp cut-off at short wavelengths. [1]

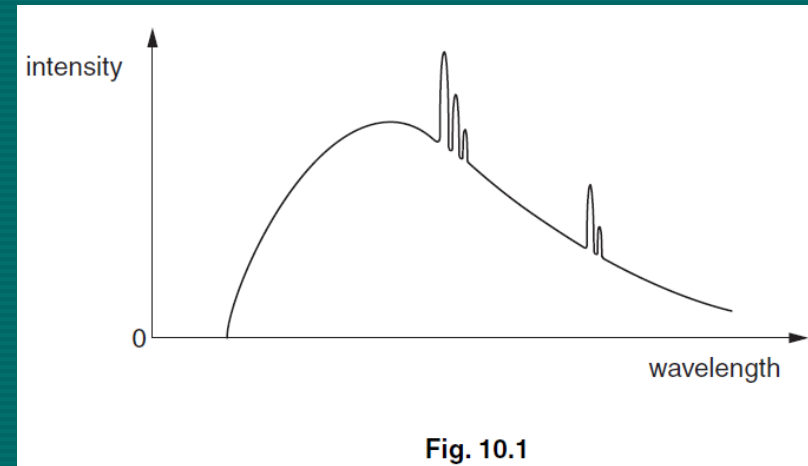


Fig. 10.1

Solution:

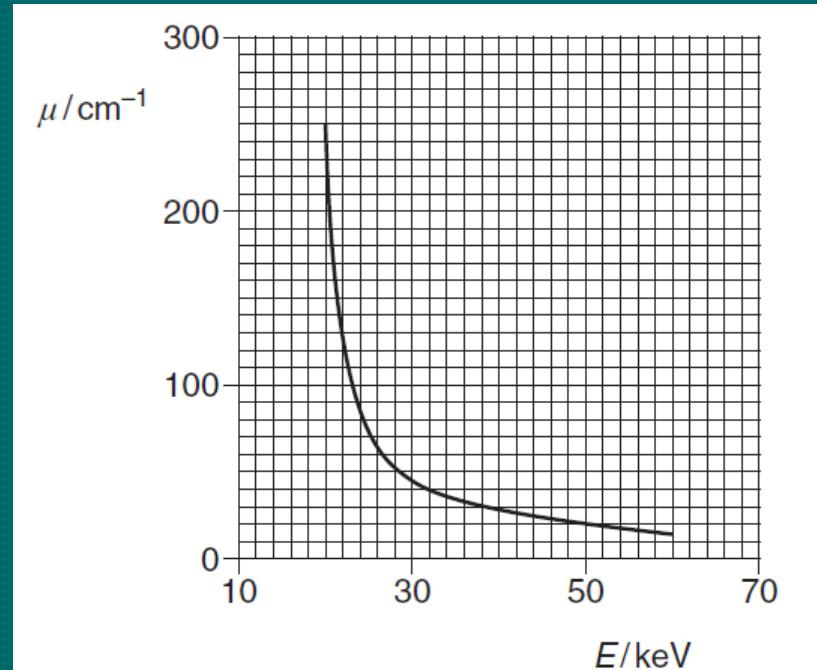
- (a) (i) e.m. radiation / photons is produced whenever a charged particle is acceleratedM1
wavelength depends on magnitude of accelerationA1
electrons have a distribution of accelerationsA1
so continuous spectrumA0 [3]
- (ii) *either* when electron loses all its energy in one collision
or when energy of electron produces a single photonB1 [1]

X-Ray (P42-Nov 2009) (2/2)

(b) The variation with photon energy E of the linear absorption coefficient of X-rays in soft tissue is illustrated in Fig. 10.2.

(i) Explain what is meant by *linear absorption coefficient* [3]

(ii) For one particular application of X-ray imaging, electrons in the X-ray tube are accelerated through a potential difference of 50 kV. Use Fig. 10.2 to explain why it is advantageous to filter out low-energy photons from the X-ray beam. [3]



Solution:

- (b) (i) parallel beam (in matter) B1
 $I = I_0 \exp(-\mu x)$ M1
 I , I_0 , (μ) and x explained A1 [3]
- (ii) *either* low-energy photons absorbed (much) more readily B1
or low-energy photons (far) less penetrating B1
low-energy photons do not contribute to X-ray image B1
low energy photons could cause tissue damage B1 [3]

X-Ray (P4-June 2007) (2/2)

- (a) Explain the principles behind the use of X-rays for imaging internal body structures.
- (b) Describe how the image produced during CT scanning differs from that produced by X-ray imaging.

Solution:

- | | |
|--|----|
| (a) X-ray beam directed through body onto detector (plate) | B1 |
| different tissues absorb/attenuate beam by different amounts | B1 |
| giving 'shadow' image of structures | B1 |
| any other detail e.g. comment re sharpness or contrast | B1 |
|
 | |
| (b) X-ray image is flat OR 2-dimensional (1) | |
| CT scan takes many images of a slice at different angles (1) | |
| these build up an image of a slice through the body (1) | |
| series of images of slices is made (1) | |
| so that 3D image can be built up (1) | |
| image can then be rotated (1) | |
| <i>1 mark for each point, max 5</i> | B5 |

30.8 Positron Emission Tomography

Positron Emission Tomography or PET scanning is another tool in the diagnostic toolbox of modern medicine. It has a range of uses: investigating, diagnosing and monitoring treatment of cancers, heart disease, gastrointestinal disorders and brain function.

The principle operation of PET is different from CT and ultrasound scanning. CT and ultrasound look at the patient from the outside, whereas PET looks at the patient from the inside. A small amount of tracer, sometimes referred to as a **radiotracer**, is injected into a vein, travels round the body and is absorbed by organs and tissues. It is the radiation from this that is used to produce the image.

Radiotracers

There are several different radiotracers used in PET, an example being a glucose based molecule, onto which a radioactive-nuclide, fluorine-18, is attached. This substance is known as fluorodeoxyglucose. The fluorine-18 nuclide decays by emitting a β^+ -particle, a positron. The advantage of using a glucose-based tracer is that it is taken up at different rates by different tissues or organs. Cancer cells are more metabolically active than surrounding healthy cells, consequently they absorb glucose at a higher rate and thus emit radiation at a greater rate. This will then appear on the screen as a bright area, allowing doctors to identify diseases and also determine the progress and effectiveness of any treatment used for the disease. PET scans are not only used for the detection of cancers but are a diagnostic tool in investigating blood flow, heart disease and brain injuries, and they are also being used to investigate Alzheimer's disease and other forms of dementia.

PET scans are unique in that they are able to pinpoint molecular activity within the patient's body, rather than looking at the body from outside. Consequently, they can identify disease in its earliest stages, meaning that there is a greater chance of successful treatment. They can also be used to track a patient's immediate and ongoing response to treatments.

What happens in positron emission?

PET scanners require a radioactive isotope that decays by β^+ emission, the emission of a positron, the antiparticle of the electron, which you met in [Chapter 15](#). Most β^+ emitters are not naturally occurring isotopes and are made by firing protons at target nuclei.

The positron moves through the patient's tissue and within a very short distance (significantly less than a millimetre) it will encounter an electron. The pair will annihilate and their mass becomes pure energy in the form of two γ -rays that move apart in opposite directions. The concept of mass-energy is discussed in detail in [Chapter 29](#).

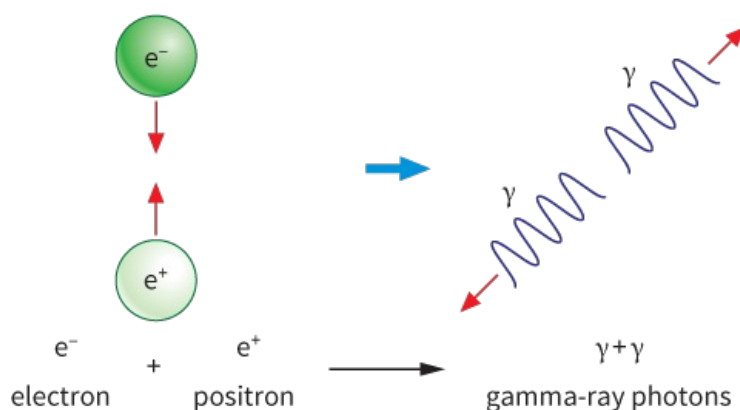


Figure 30.20: Energy is released in the annihilation of a positron and an electron.

In the annihilation process, as in all collisions, both mass-energy and momentum are conserved. The initial kinetic energy of the positron is small – negligible compared to their rest mass-energy – hence, the γ -ray photons have a specific energy and a specific frequency that are determined, solely, by the mass-energy of the positron-electron pair.

The energy of a photon is given by:

$$E = hf$$

where h is Planck's constant, and f is the frequency of the photon.

The momentum of a photon is given by:

$$P = \frac{E}{c}$$

where c is the speed of electromagnetic radiation in a vacuum.

KEY EQUATION

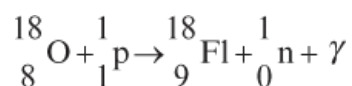
$$\text{energy of a photon} = E = hf$$

$$\text{momentum of a photon} = p = \frac{E}{c}$$

The production of suitable radioisotopes

You will remember from your work on magnetic fields that a charged particle entering a magnetic field at right angles to the field will travel in a circular path. The cyclotron works on this principle, however, the particles are continuously accelerated by an alternating electric field as they go round the circle, thus they travel in a spiral path before they are released and collide with the target nuclei. Figure 30.21 shows an early cyclotron. The principle of the cyclotron is shown in Figure 30.22.

In the production of fluorine-18, oxygen-18 nuclei are bombarded with protons and the following reaction takes place:



The isotope, fluorine-18, has a half-life of just under two hours. This means the patient is not subjected to radiation for a long period of time. However, it also means that the radiotracer needs to be made up freshly, probably on site, to be most effective.

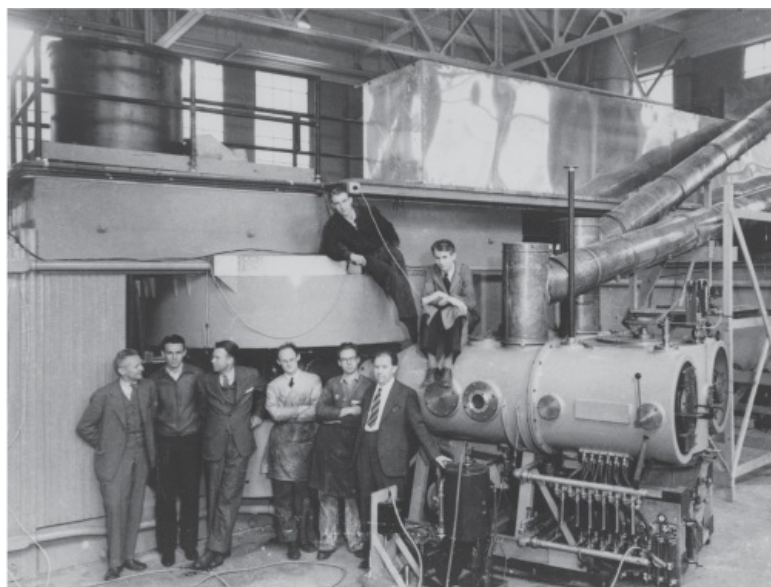


Figure 30.21: The cyclotron at the Lawrence Radiation Laboratory, Berkeley, soon after completion in 1939.

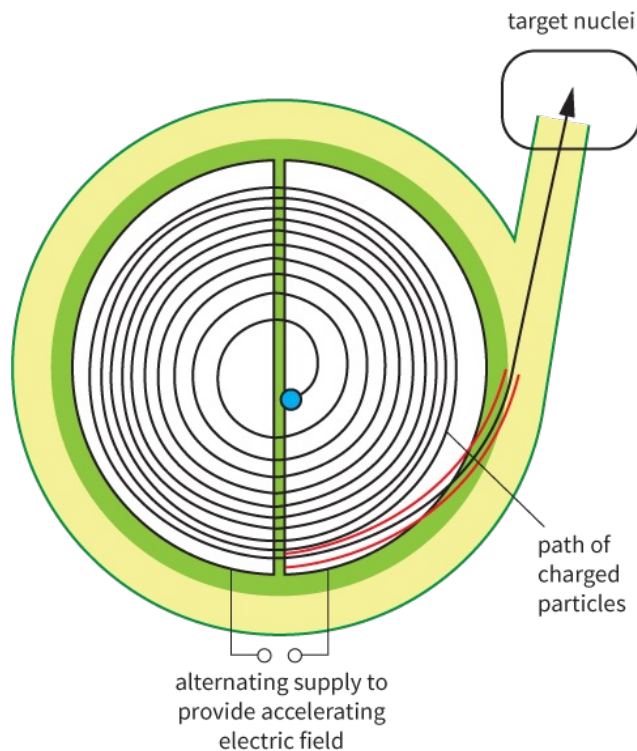


Figure 30.22: The principle of the cyclotron. Note, the magnetic field is into the plane of the page.

Questions

- 18** Suggest the reason why, in PET scanning, it is important that the positron meets an electron within a very short distance from its point of emission.
- 19** Explain why the γ -rays produced in positron-electron annihilation must travel at 180° to each other.
- 20** Fluorine-18 decays by β^+ emission. Write a nuclear equation to show this decay.
- 21 a** Calculate the energy released when a positron and an electron annihilate.
(Mass of an electron = mass of a positron = 9.1×10^{-31} kg.)
- b** Calculate the frequency of the γ -rays emitted.
- c** Calculate the momentum of the one of the γ -rays emitted.

Detecting the γ -rays

The patient being scanned is placed on a bed with a series of rings of detectors, in a donut type shape. The patient on the bed is moved through the detectors, so that a series of images of 'slices' through the patient are made in similar manner to those made by a CT scan. Indeed, PET scans are often combined with CT scans so that more information is gathered.



Figure 30.23: A patient being prepared for a PET/CT scan by a radiologist. Note the donut shaped ring,

which contains the detectors and through which the patient will be moved.

The detectors of the γ -ray photons consist of two parts: a crystal that scintillates and a photomultiplier. When a high energy γ -ray photon is incident on the crystal, an electron is excited into a very high energy state. As the electron travels through the crystal, it loses energy and excites more electrons; these electrons then decay back to their original state, emitting visible light photons. The photons produced by the scintillator are then converted into an electrical signal by the photomultiplier tube – these signals are then fed to a computer that can plot back where the photon pair was originally produced.

Reconstruction of the image

Figure 30.24 shows a simplified view of the detectors in a PET scanners. They form a series of rings around the patient. The γ -ray photons, formed by an electron-positron annihilation, travel from a point very near to the event. They travel in a straight line and in opposite directions and strike the detectors as shown. A line (known as the **line of response**) can be drawn, joining the two detectors. Using the time lapse between the two photons arriving at the detectors, the position on the line of response can be established. In practice, there are many annihilations and sophisticated computers analyse the data and convert it into an image. The numbers of photons arriving from a particular point determine the concentration of the tracer at that point. Where there are many arriving per unit time, it means that there is a high concentration of tracer and this will appear as a bright point on the image.

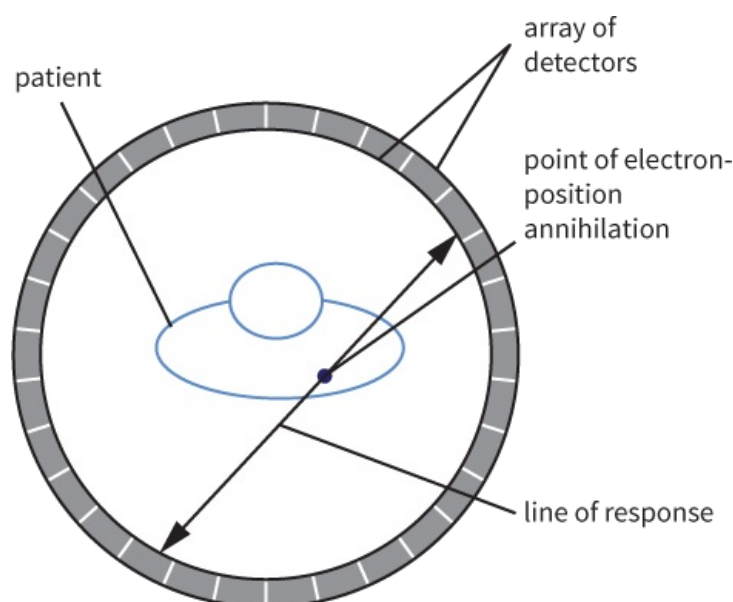
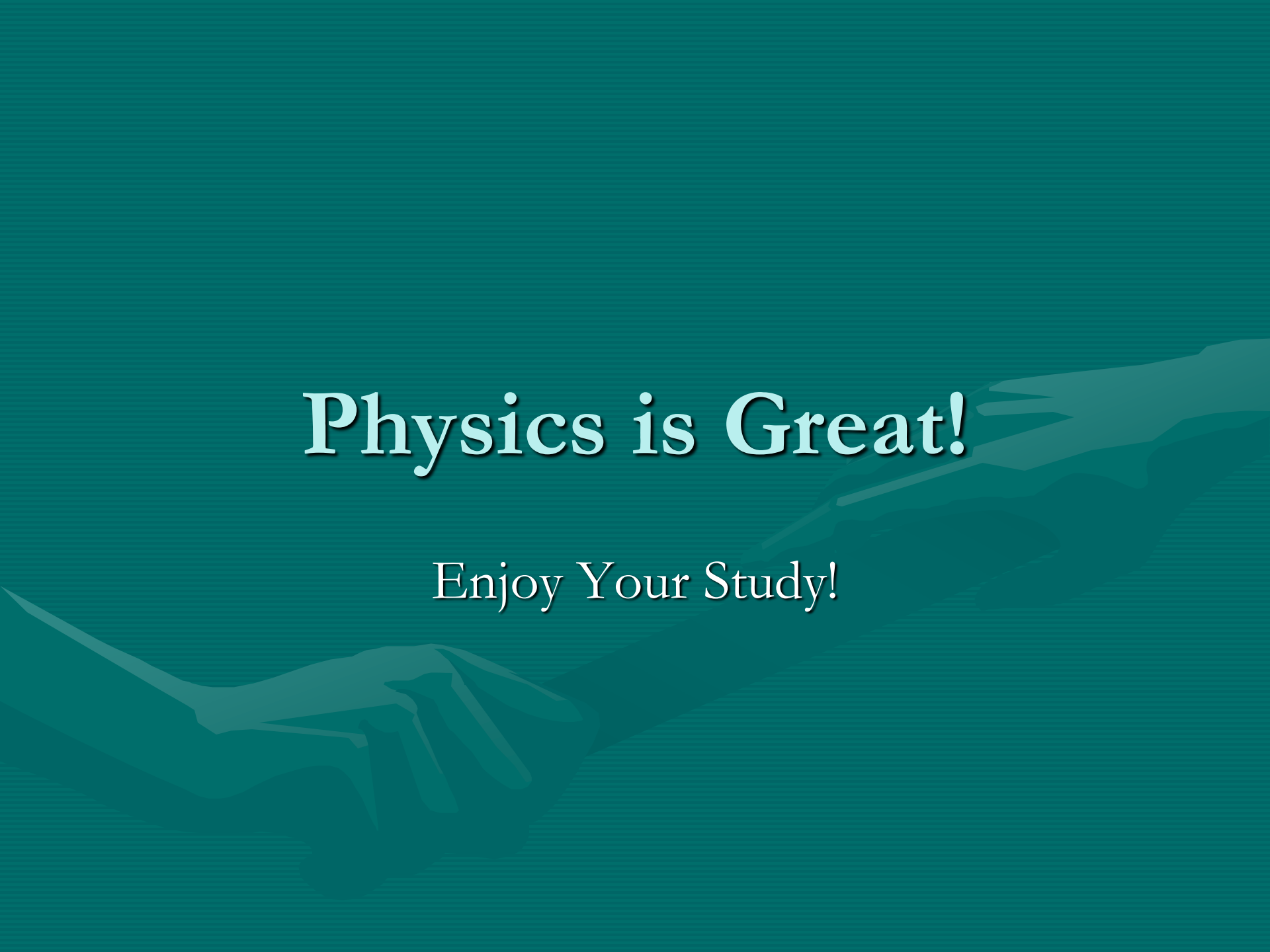


Figure 30.24: The arrangement of detectors in a PET scanner.

REFLECTION

It is about 120 years since X-rays were discovered. Modern medicine has many methods for looking inside the bodies of people who are unwell or have suffered injuries. Use the internet to find as many different methods as you can. Try and draw a timeline to show when these methods were developed.

What did you learn about yourself as you worked on this activity? Did you find it a useful way of learning?

The background of the slide features a stylized, semi-transparent illustration of two hands shaking, symbolizing agreement or partnership. The hands are rendered in a light teal color, contrasting with the darker teal background. The handshake is positioned diagonally across the lower half of the slide.

Physics is Great!

Enjoy Your Study!